A mode-coupling theory of phonological organization

Sam Tilsen
Department of Linguistics, Cornell University
tilsen@cornell.edu

Abstract

Phonological theories provide macroscopic descriptions of order in speech, but have not given much attention to how those descriptions might be derived from the relatively microscopic dynamics of the nervous system. This paper proposes a microscopically inspired theory of phonological organization in which the systems that govern articulatory actions (gesturals) are composed of internal subsystems (gestural excitations). In the preparation of speech, gestural excitations can transiently couple (phase- and frequency-lock) to a generic set of oscillatory coordination modes. In the execution of speech, a feedback-regulated competitive selection process gives coordination modes heightened activation, causing coupled gestural excitations to exert strong influences on control of the vocal tract. Phonological organization in speech preparation is conceptualized as a stochastic process in which gestural excitations couple to coordination modes, governed by probabilities derived from hypothesized constraints. The framework is used to interpret segmental, moraic, syllabic, metrical, and prosodic word-related patterns.

1 Introduction

The acoustic pressure field of human speech is highly ordered. For example, when an acoustic state associated with a consonantal sound occurs, our uncertainty about the acoustic state a tenth of a second later is reduced: the later state becomes more likely to be associated with a vocalic sound. In general, the speech production system produces far less information in its surroundings than the amount of information that it logically could produce. This can be readily observed in distributions of short-time spectral representations of the acoustic waveform, and in distributions of vocal tract states quantified through articulatory imaging. Limitations on information produced in speech are a consequence of not only mechanical constraints but also—and more interesting to phonologists—constraints on the cognitive system that interacts with the acoustic and articulatory environment. From this perspective, one aim of theoretical phonology is to understand how constraints on cognitive systems result in acoustic/articulatory order across a range of spatial and temporal scales.

This manuscript presents a novel model of a particular part of this system and its constraints, as an instantiation of *mode-coupling theory*. Mode-coupling theory proposes (i) that the systems responsible for controlling vocal tract states (*gesturals*) are composed internally of subsystems (*gestural excitation modes*), and (ii) that these internal subsystems can transiently couple to highly generic systems (*coordination modes*) via a frequency-adaptation mechanism. The theory makes use of several

previously developed mechanisms to account for how the coupled states of these two types of systems drive movement.

In the remainder of this section, we elaborate on the motivation and aims of mode-coupling theory. Section 2 presents the model in detail. Sections 3 and 4 provide interpretations of various phonological patterns and a general discussion. All code for simulations is documented in https://github.com/tilsen/mode-coupling, including detailed descriptions of the previously developed mechanisms (competitive selection and vocal tract field control), which are too lengthy to include in this manuscript.

1.1 Phonological theory as macroscopic description of pattern

From an information-theoretic perspective, phonological PATTERNS are always associated with joint distributions of acoustic and/or articulatory environmental states that are more ordered (lower in entropy) than they logically need to be. Phonological theories construct vocabularies (systems of conceptual metaphors and image schemas) for understanding the causes of that orderedness, particularly those causes that are associated with processes in the human central nervous system, i.e. cognition. In order to usefully attribute causes, we need clarity on how to define the cognitive systems that give rise to environmental states, and we need to be clever about how we define the environmental states themselves. For current purposes, we refer to the collection of systems outside of the central nervous system (i.e. the acoustic pressure field; vocal organs; and the sensorimotor periphery) as the environment (or *surroundings*) of the cognitive system.

Imagine a very powerful observer, who continually and simultaneously records a detailed, high-dimensional record of the contribution of every human to their acoustic environment, along with a high-dimensional record of the articulatory/somatic environment, i.e., the mechanical state of the vocal system. The observer may observe a system state x_t and at a later time x_{t+1}, or they may observe a state x associated with some particular sub-space and at the same time another state x' associated with some other sub-space. For generality, we refer to states x and y to cover both the spatial and temporal cases. If we think of states x and y as random variables, then the information *produced* by observing x and y is equivalent to the reduction of uncertainty associated with the observation. We quantify this information as a reduction of entropy (Shannon, 1948): prior to the observation, uncertainty was $H(x,y) = -\sum_{ij} p(x_i, y_j) \log p(x_i, y_j)$, i.e., the average value of the surprisal, $-\log p(x_i, y_j)$. Subsequent to the observation, uncertainty is 0. Hence the information produced is the reduction in uncertainty, equivalent to the entropy before observation.

In the case where the observer has already observed x but not yet y, the information produced by observing y is the conditional entropy $H(y|x)$, which is the expected value of the conditional surprisal, $-\log p(y|x) = \log p(x) - \log p(x,y)$. When x and y are statistically independent, the information produced by observing y (i.e. the reduction of conditional entropy) is maximized, because $H(y|x) = H(y)$. In the dependent case, $H(y|x) < H(y)$ and $H(x,y) < H(x) + H(y)$, so less information is produced by observing y than in the independent case. The dependent circumstance is a form of PATTERN, and we can say there is ORDER in the joint distribution $p(x,y)$ in the sense that its entropy is lower than the highest entropy possible given its marginal distributions, $H(x) + H(y)$. Put differently, patterns are always associated with a spatial and/or temporal statistical dependence of state variables.

Many patterns are not under the purview of phonological theory. Let us imagine that the observer makes observations of x and y that are very proximal in space and time (e.g. the intralabial acoustic pressure field of a particular speaker, observed several milliseconds apart). There is a very large amount of order in the joint distribution of these states. But this orderedness is a consequence of mechanical constraints on changes in vocal tract state (conservation of energy and momentum, i.e., microscopic time- and space-translation symmetry applied to microscopic degrees of freedom). Such order is generally not the concern of phonology.

Instead let us imagine that the observer makes acoustic field observations 100 ms apart. Once again there is substantial order in the joint distribution of the state variables, but this order can no longer be attributed primarily to mechanical conservation laws. Given (i) that the acoustic field is determined by the geometry of the vocal tract and its aerodynamic pressure gradients, (ii) that vocal tract geometry is determined by movements of articulators resulting from muscle fiber contraction, (iii) that said contraction is governed by neurons in the brainstem motor nuclei, and (iv) these brainstem motorneurons interact with neurons in a variety of cortical and subcortical structures, it can plausibly be inferred that the central nervous system is responsible for the observed order.

Unfortunately, not only the environmental state but also the nervous system state is very high dimensional. We are limited in our ability to reason about patterns in such high-dimensional spaces, and particularly for the central nervous system, we are limited in our technological capacity to empirically observe and analyze the high-dimensional state. Consequently, for the sake of reasoning about order, it is a practical necessity to construct macroscopic descriptions of the relevant states. The observer must strategically define the macroscopic state as a function of the microscopic one (or at least imagine that such a function exists).

In mainstream phonological theories, the environment is the acoustic field (Fig. 1 (a)), and the macroscopic environmental states are entities called segments (Fig. 1 (b)). The premise of segmental reasoning is that non-overlapping periods of time in the acoustic signal are instances of segment types, which have identifying properties. Most phonology students are at some point taught to temporally partition records of the acoustic field in this way through visual inspection of waveforms and spectrograms. It is not hard to imagine that there is a function from the microscopic acoustic field states to these types, but attempts to elaborate this function not have been very successful, and there are a number of reasons to suspect that the basic premise is misguided.

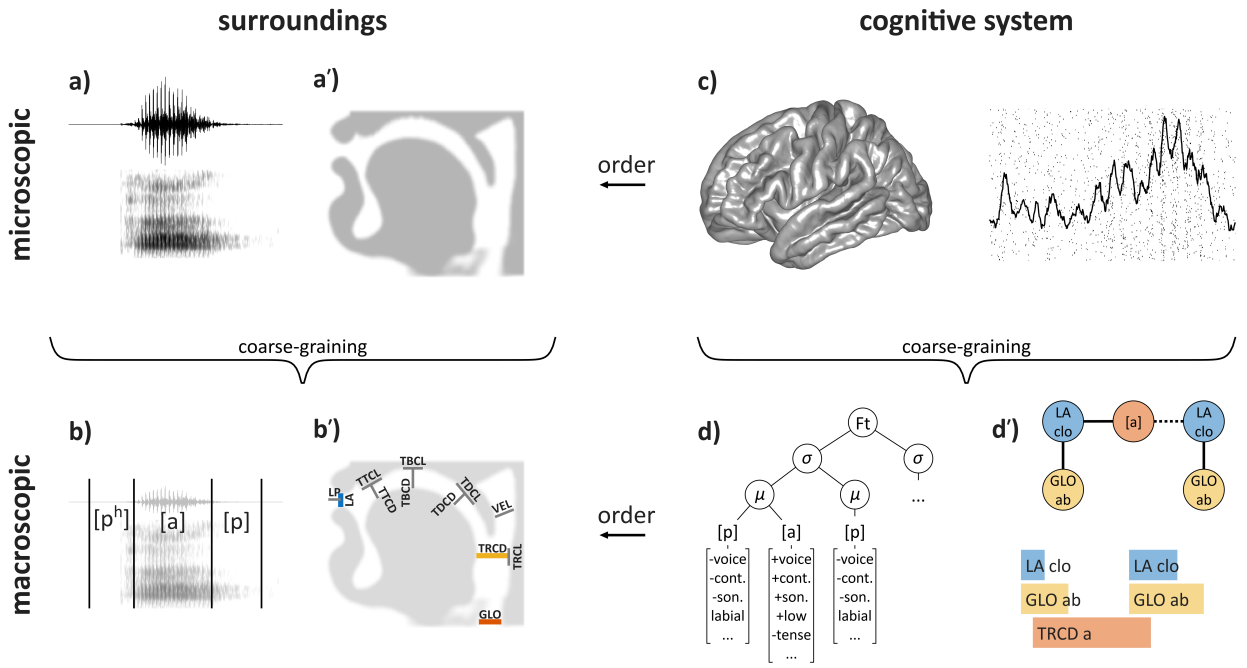


Figure 1: Phonological theories aim to account for how states of the central nervous system (c) give rise to order in the acoustic and articulatory surroundings (a, a′), defined macroscopically (b, b′). Object-based theories (d) and gestural theories (d′) are alternative macroscopic descriptions of central nervous system states.

Nonetheless, phonological theories have transferred segmental reasoning about the acoustic surroundings to a conceptual model of the cognitive system, where segments (or *phones* or *phonemes*) are object-like entities that contain *features* (which account for their identifying properties), and where segmental entities are related through a hierarchical structure of additional hypothetical entities (*syllables*, *moras*, *feet*, etc.), as shown in (Fig. 1 (d)). The transfer of segmental reasoning from analysis of the acoustic surroundings to a conceptual model of cognition is plausibly a consequence of the fact that most early linguistic theory was done by lexicographers using phonographic writing systems, along with the circumstance that scientific knowledge is predominantly communicated via the technology of writing itself. See Faber, 1992; Linell, 1988, 2005; Ong, 2013; Port and Leary, 2005 for similar critiques.

In an alternative form of phonological reasoning, *gestural phonology*, the environmental focus is the time-varying state of the vocal system (Fig. 1 (a′)), analyzed macroscopically (Fig. 1 (b′)) as states (often apertures) associated with vocal organs (lips, tongue, larynx, lungs, etc). In the gestural framework of Articulatory Phonology / Task Dynamics (Browman & Goldstein, 1986; Saltzman & Munhall, 1989), the hypothesized macroscopic cognitive systems are entities called *gestures*. When gestural systems transition to an active state, they change the equilibria of systems that control the vocal tract more directly, referred to here as *vocal tract control systems*. Macroscopic analyses in this framework often employ gestural *activation intervals* and *coupling graphs* (Fig. 1 (d′)). Coupling graphs represent phase coupling between oscillatory systems that have been hypothesized to govern when gestures become ac-

tive. The activation intervals indicate periods of time when gestural systems influence vocal tract control systems.

Hence in the status quo of phonological theory, there are primarily two approaches, with fundamentally different vocabularies for macroscopic description of the environment and the nervous system. Segmental/symbolic theories focus primarily on the acoustic environment and construct object-like entities to describe cognitive states; gestural theories focus primarily on the articulatory environment and describe cognitive states in terms of systems that drive movement. Critically, in neither approach has much attention been given to the function that derives the hypothesized cognitive states from the microscopic, high-dimensional state of the nervous system (Fig. 1 (c)). This lack of attention is quite understandable, because our current technological capacity to observe the nervous system state is limited. Nonetheless, it may be of value to speculate upon such derivation in the construction of the macroscopic description.

Any macroscopic description, however useful, can only withstand scrutiny as long as it is not shown to be microscopically *incompatible*: the extent to which the emergent macroscopic dynamics can be understood as the consequence of microscopic dynamics is an important criterion for the validity of the description. The swarming behavior of a flock of birds and the magnetization of bar of iron are not magic; these phenomena arise from the interactions of the individual sub-systems that comprise the whole, regardless of whether we understand the microscopic details of these subsystems and their interactions. In Marrian terms, computational and implementational descriptions are not independent, and our ultimate goal should be to relate them (Marr, 1982; Marr & Poggio, 1977).

1.2 Theoretical desiderata

From a biophysical perspective, a macroscopic theory of phonological organization should be consistent with, or at least should not be inconsistent with, our understanding of both the microscopic dynamics of the nervous system and the microscopic dynamics of acoustic/articulatory environment. Below we discuss some ways in which inconsistency can be avoided:

Empiricality. First, the theory must aim for empirical adequacy, coverage, and unification. Its goal should be a macroscopic description of phonological cognition that provides mutually coherent mechanisms for understanding a wide range of empirical patterns in the environment. These include within-language regularities in the patterns of sound/action that occur contemporaneously (*phonotactics*), the range of possibilities for such patterns across languages or speech communities (*typology*), temporal variation in actions themselves, especially variation that is systematically predictable (*articulatory timing*), developmental changes in the organization of action (*acquisition*), and possibilities for atypical patterns arising from less probable state space trajectories (*phonological errors*).

Non-spatialization of time. Second, the theory should avoid metaphors for cognition in which (i) actions are objects and (ii) temporal order is a spatial arrangement of objects. Standard object-based representations such as those in Fig. 1 (d) entail that spatial arrangement (by convention, left-to-right order) maps to temporal

order. These representations conceptualize variation in temporal patterning with operations on schemas of hierarchically arranged object-like phonological units (e.g. linking and de-linking, copying or deletion, spatial relocation, etc.). Schemas of this sort treat what should be viewed as state space *trajectories* as if they were states *per se*. Hence the schemas are somewhat incoherent as representations of state, despite being pervasively conceptualized as such.

Finiteness. The number of state space dimensions should be fixed, for a given speaker, on the typical timescales of utterances. The description cannot arbitrarily proliferate subsystems and corresponding state space dimensions, just when needed. In object-based structural representation of an action sequence in which some action is repeated, such as bilabial closure [p] in Fig. 1 (d), the representation effectively treats the second instance of [p] as if it is a separate entity from the first, implying that there must be an additional subset of state space dimensions for that instance. This begs the question of how the duplication of state space dimensions translates to a state of the nervous system. No one would claim that the neural populations that drive motor actions are "copied" whenever a sequence of actions contains identical elements, and yet, this is effectively what the schematic/symbolic representation implies, because it allows the same entity to occur in multiple places in the schema. The ad-hoc replication of entities has been called the problem of *multiplicity* in previous work (Tilsen, 2019c) and has been discussed as a matter of type vs. token representation (Jordan, 1997). We should require that cognitive states be states *of systems* (types) not *trajectories of system states*, i.e. the temporal outputs of systems (tokens).

Memorylessness (first-order Markovian dynamics). In a Markovian model the probability of the system being in a particular state at a particular time depends on prior states over a finite period of time, i.e. $P(X_{t+1} \mid X_t, \ldots, X_{t-k})$ does not depend on states before X_{t-k}. In a first-order Markovian model, the next state depends only on the current state, i.e. $P(X_{t+1} \mid X_t, X_{t-1}, \ldots) = P(X_{t+1} \mid X_t)$; for higher order Markovian models, the system can be said to have a "memory" of past states. All fundamental physical dynamical laws are first-order Markovian with appropriate definition of state, but macroscopic/coarse-grained laws need not be. Whenever a model of a physical system possesses memory, it is because the macroscopic description has been constructed so as to relocate some microscopic degrees of freedom to an ad hoc subspace—memory.

A more physically appropriate coarse-graining strategy in such circumstances is to model memory as a slower timescale macroscopic dynamic of the system. To give an example, the compression that occurs when one lays their head on a memory foam pillow might appear to depend on its history, but not because the state of the pillow includes a record of past states. Several additional state variables, such as the configuration of the polymer network, temperature, and air density, evolve more slowly than the load applied by a head. The "memory" that we refer to when describing cognitive systems is not fundamentally different. What appears to be dependence on past states (i.e. memory) is more coherently understood as a dependence of particular focused sub-system states on ignored sub-systems with relatively slow dynamics. Specifically, much of the "memory" associated with lexical knowledge can be understood as coarse-grained description of synaptic efficacy and neuronal connectivity, which are convenient to model on slower timescales.

Non-autonomous dynamics. The nervous system, a far-from-equilibrium open system,

interacts strongly with its surroundings. First and foremost of these is the metabolism that provides energy (in the form of electrochemical gradients and molecular configurations) for cycles of neuronal polarization and depolarization. On larger spatial and temporal scales, order in environmental acoustic fields—speech produced by other speakers—is another hefty influence. Whenever we choose to model a particular subsystem of the universe (our focused system), we make an analytical choice to simplify our model of the complement of that subsystem, i.e. the remainder of the universe (the surroundings).

Short of ignoring the surroundings entirely, we may attempt to incorporate some of its influences on the focused system as time-dependent terms in the dynamical law for the focused system. Abstractly, we might start with $\dot{x} = f(x, t)$, as a practical necessity. Later, we might expand the scope of the model, i.e. $\dot{x} = f(x, y), \dot{y} = g(y, t)$, relocating time-dependence (i.e. non-autonomy) to a different, typically smaller portion of the universe. All physical systems, except perhaps the universe as a whole, are inevitably non-autonomous in this sense. For an ideal gas in perfectly rigid, perfectly insulated box, no matter or energy crosses the boundary between the box and its surroundings; in real boxes, however, walls are neither perfectly rigid nor perfectly insulating: some energy and matter will eventually transfer between the system and its surroundings. The "box" of the cognitive system has little insulation and is hardly rigid: metabolic and other surroundings influences are fast and strong. Macroscopic descriptions should therefore embrace non-autonomy by allowing themselves to incorporate influences from exogenous systems.

Microscopic compatibility. Theoretical descriptions of speech production should not be incompatible with our current understanding of the nervous system. They should not preclude the construction of a bridge between the fine-grained spatial and temporal scales of analysis that are relevant to describing the nervous system and the coarse-grained scales that are more useful for describing phonological patterns. This desideratum is often disregarded in contemporary theories by appealing to the primacy of a "computational" level of description, as discussed by Marr (1982) and Marr and Poggio (1977). Such appeals misinterpret Marr, who, despite emphasizing the importance of the computational level, also repeatedly advocated for integrating the computational (relatively macroscopic) level with lower-level (relatively microscopic) algorithmic and implementational ones (see Tilsen, 2019c, pp. 264–271 for extended discussion and examples).

Compatibility relates to the philosophical issue of strong vs. weak emergence. Relatively macroscopic patterns can be viewed to emerge from microscopic ones, where *emerge* connotes that the macroscopic dynamics are readily deducible/derivable from the microscopic dynamics. In strong emergence, the disconnect between scales is viewed as an ontological necessity: macroscopic behavior is *a priori* not reducible to microscopic behavior. Alternatively, weak emergence recognizes that our inability to derive the macro from the micro is due to our ignorance of the latter, and ultimately the complexity thereof. Strong emergence is philosophically untenable in my opinion: it creates a dual space for analysis of causes, one that will never be accountable to what can be physically observed. Weak emergence, in contrast, allows for the possibility that, with diminished ignorance in the future, we could ultimately derive phonological patterns from neural ones.

What is not possible at the current time is to relate neural and cognitive scales

of analysis in an empirically testable way—in many cases we can provide only speculative ideas regarding how macroscopic systems may be microscopically derived. Because these are often phenomenologically motivated, there is inherent subjectivity in whether they are plausible. Nonetheless, it is preferable to have *some* idea in mind (however vague that may be) of how the the derivation could be accomplished. The desideratum of compatibility is in effect a meta-rationale for physically motivated desiderata: non-spatialization of time, finiteness, memorylessness, and non-autonomy. These are corollaries of the insistence that a macroscopic theory should not be aphysical, because the nervous system is unarguably a physical system.

1.3 Heuristics of mode-coupling theory

Mode-coupling theory is an attempt to lay the groundwork for a bridge between microscopic and macroscopic descriptions, to demystify emergence. The first premise of the theory is that some of the order associated with speech is a consequence of constraints on system states/state trajectories, rather than operations on structures of objects. Spatial relations play a very different role, compared to the segmental/symbolic vocabulary: they *never* encode temporal patterns by using the metaphor that a temporal order of events is a spatial arrangement of objects.

The second premise is that, microscopically, the relevant systems are populations of neurons (most likely in posterior premotor cortex, but also interacting with populations in other cortical areas and with various subcortical systems). Moreover, these populations exhibit collective oscillations that motivate a lower-dimensional, macroscopic description of interactions based on phase-/frequency-locking (coupling of oscillatory systems). Here these systems are called *gesturals*. The term is intended to evoke the concept of a *gesture*, first developed in the theoretical frameworks of Task dynamics and Articulatory phonology (Browman & Goldstein, 1986; Kelso et al., 1986).

The third premise is that gesturals, through their interactions with neural populations in other areas, govern the vocal tract system state. Gesturals are differentiated on the basis of cortical topographies—the tonotopy of auditory cortex and the somatotopy of primary motor and somatosensory cortices, both of which are consequences of phylogenetic and ontogenetic characteristics of the human species. The premises regarding collective oscillation and a topographic basis for differentiation are somewhat loosely empirically supported in specific application to articulatory control, but in general the premises are well-established and it is not hard to envision how future advances in neural imaging will make disconfirmation possible.

The last premise, perhaps the most radical one, is that gesturals can be analyzed as comprised of internal subsystems—*gestural excitation modes*—whose frequencies can adapt to the frequencies of host systems called *coordination modes* or μ-systems. Mode-coupling theory holds that some of the environmental information produced by motor action and its concomitant acoustic order can be usefully understood as an indirect consequence of constraints on how gestural excitation modes couple to coordination modes. In effect, coordination modes make available a finite set of state space attractors for gestural excitations, and hence are a large source of phonological ORDER.

Before presenting a specific macroscopic model, let us motivate the phenomenology of mode-coupling theory by considering how its macroscopic concepts might be

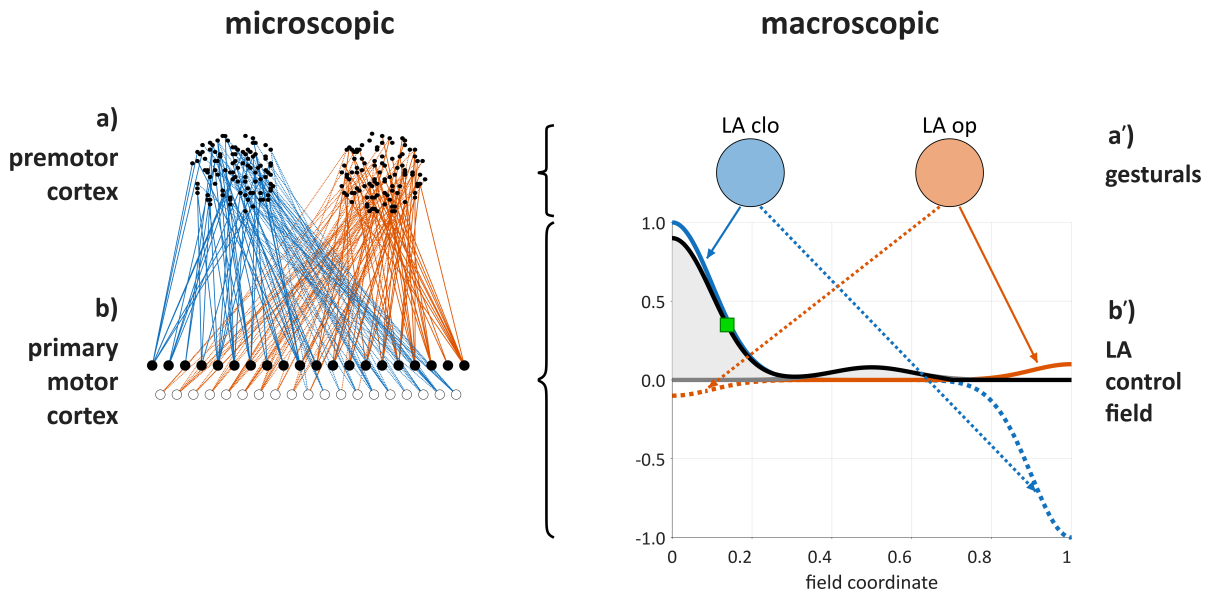


Figure 2: Microscopic interpretation of model abstractions. (a) premotor subpopulations have different spatial patterns of projection to excitatory and inhibitory neurons in somatotopically organized populations in primary motor cortex (b). Macroscopic/abstract interpretations: different gesturals (a') have different patterns of excitatory and inhibitory influence (solid and dashed lines) on the activation (black line) of a vocal tract state control field (b'), here lip aperture (LA). The current equilibrium state of lip aperture is the centroid of the activation (green square).

grounded microscopically—in other words, we address how the heuristic abstractions of the theory conform to the desideratum of microscopic compatibility.

Somatotopic organization. Assume that primary motor cortex is somatotopically organized such that there are subpopulations of neurons associated with geometrically/functionally defined vocal tract state dimensions (Fig. 2 (b)), such as lip aperture, tongue tip constriction degree/location, tongue body constriction degree/location, velum aperture, glottal aperture, etc. Similarly, assume that there is a spatial organization of subpopulations in primary somatosensory cortex that maps in a highly systematic way to the vocal tract state-related organization in primary motor cortex. There is plenty of evidence that gross somatotopies exist in these areas, along with some evidence that more specific speech articulation-related somatotopies exist (Bouchard et al., 2013; Chartier et al., 2018; Conant et al., 2014; Conant et al., 2018). Exactly how these somatotopies are instantiated and what macroscopic description is most appropriate for them have yet to be determined. Relatedly, various studies of saccadic and manual motor control have suggested that targets in the visual field and peripersonal space are reflected in topographically organized neural populations (Graziano et al., 2002; Robinson, 1972).

Somatotopy in primary motor and somatosensory cortices motivates a macroscopic scalar field model (Fig. 2 (b')). The activation at each field coordinate is microscopically a short-term integration of layer V pyramidal depolarization-rate

in a local region of the population. The activation field determines the current equilibrium of some dimension of the vocal tract state; microscopically this is due to synaptic projections from primary motor cortex to brainstem motorneuron nuclei that control muscle tension. The centroid of activation (green square) in the field serves as the control target for the corresponding vocal tract state dimension.

Gestural influences on tract fields. Gesturals are macroscopic descriptions of the states of premotor cortical neural populations, which are differentiated on the basis of their patterns of synaptic projection to neurons in primary motor and somatosensory cortices. Macroscopically, this entails that each gestural is associated with some particular vocal tract state control field(s), and that gesturals can excite and inhibit activation in different regions of those fields. For example, in Fig. 2 (a') the gesturals [LA clo] and [LA op] provide different spatial distributions of excitatory and inhibitory input to the LA (lip aperture) control field: [LA clo] excites the region of the field that corresponds to bilabial closure and inhibits the region that corresponds to an open posture; vice versa for [LA op].

Collective oscillation. Gesturals are assumed to have the ability to transition to and from an active regime in which they exhibit relatively high-amplitude oscillations and an inactive regime in which their oscillation amplitudes are negligible. Microscopically these oscillations are collective oscillations of neuronal populations. There is a large body of empirical and theoretical work that supports this assumption (Buzsáki & Draguhn, 2004; Buzsáki & Wang, 2012; Singer, 2018; Wang, 2010; Wilson & Cowan, 1972) and that relates transient oscillatory activity to working memory and/or motor preparation/execution (Churchland et al., 2012; Feingold et al., 2015; Lundqvist et al., 2016). The transition to the collective oscillation regime is driven by external input and is possible because the individual neurons and/or cortical microcircuits that comprise the population exhibit a bifurcation in their dynamics between a fixed point (polarization) and a limit-cycle attractor (cycles of polarization and depolarization; Brunel, 2000; Izhikevich, 2007; Stiefel and Ermentrout, 2016).

Internal excitation modes of gesturals. One novel premise of mode-coupling theory is that gestural states can be viewed as a superposition of oscillations of a discrete set of relatively independent subsystems, referred to as *internal excitation modes*, or *gestural excitations*.

A loose physical analogy for gestural excitations is the vibration of a circular membrane, such as a drumhead. Fig. 3 (i) schematically illustrates various oscillatory modes of membrane vibration, which are characterized by the number of nodal diameters (m) and number of nodal circles (n). Nodes, where membrane displacement is zero, are shown as radial and circular lines. The presence/absence of shading indicates regions of the membrane that are above/below zero displacement in a snapshot of the vibration, and the numeric values show the relative frequency of the vibrational mode. In an idealized membrane the modes are orthogonal and evolve independently, but in real drums there are factors that couple them weakly. The distribution of energy across modes depends on the external excitation of the membrane. A skilled drummer can control the tone of a strike by varying the strike location, area, and force, and can damp modes by selectively touching the membrane. In real drums, the set of possible modes is effectively finite, because the continuum description

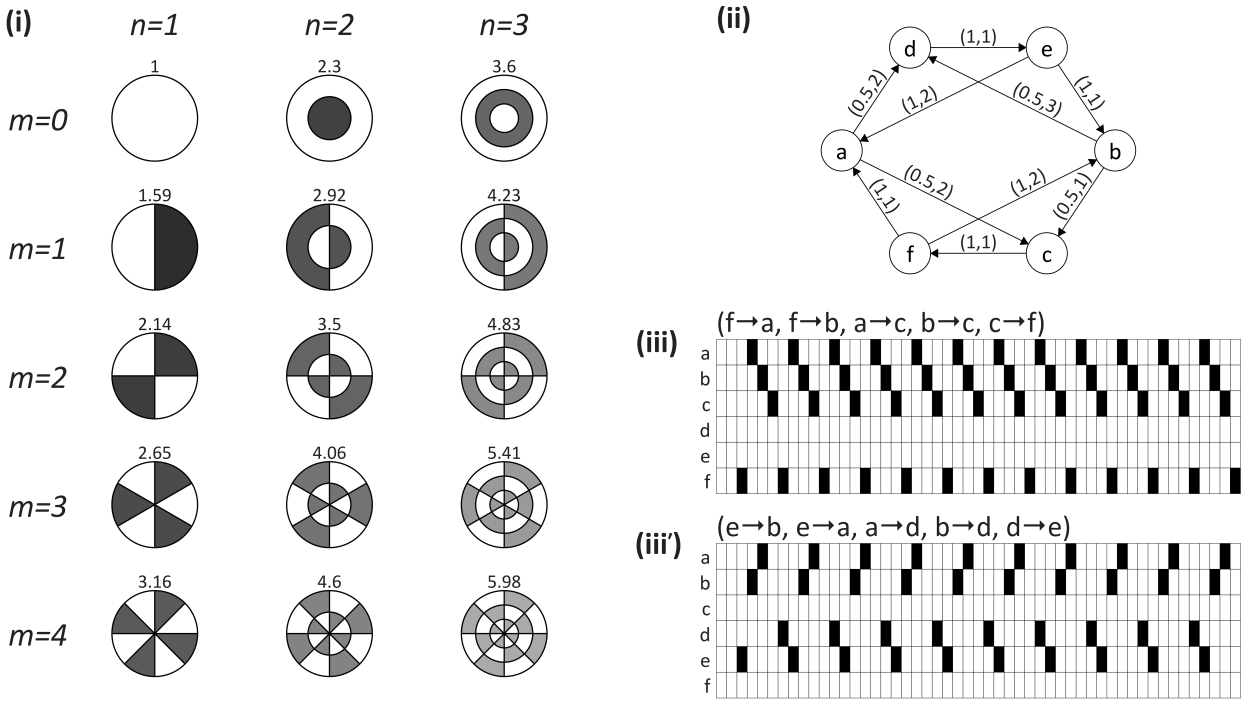


Figure 3: Excitable modes of gesturals. (i) Circular membrane vibrational modes are characterized by the number of radial and circumferential nodes. Numeric values shown are relative frequencies. (ii) Polychronous groups are a plausible basis for frequency and phase differentiation of oscillatory modes of neural populations. (iii, iii') Periodic spike patterns that result from external input to neuron *e* or *f*.

of the membrane breaks down as wavelengths approach the molecular/microstructural scale, and more practically, because higher-frequency modes are more strongly damped and require more energy to excite to a given amplitude.

The key mappings of the analogy between membranes and gesturals are: (i) the availability of relatively independent (weakly coupled) modes that correspond to periodic components of a state space trajectory; (ii) the allowance for an external system to distribute energy in different ways to different modes; and (iii) effective finiteness of modes due to physical constraints. The macroscopic state of a gestural is likewise characterized as a finite set of excitable modes, each of which has a time-varying amplitude, frequency, and phase. The drummers who excite the internal modes are external systems associated with concepts/meaning. This hypothesis is phenomenologically motivated by the problem of multiplicity: relatively independent sub-systems of gesturals are necessary to account for how it is possible that the same action may occur multiple times in the production of a form.

Regarding a microscopic basis for the existence of gestural excitations, one possibility relates to the notion that multiple cyclic polychronous groups may co-exist within a population of neurons. A polychronous group (Izhikevich, 2006; Vertes & Duke, 2010) is a set of neurons that exhibit a time-locked pattern of spikes, with the specific pattern being a consequence of (1) the topology of synaptic interaction between neurons, (2) their status as excitatory or inhibitory, and (3) axonal conduction delays. A cyclic polychronous group is one that repeats its spike pattern periodically.

A simplified illustration of two polychronous groups is shown in Fig. 3 (ii), where excitatory synapses (directed graph edges) are labeled with their weight and conduction delay. In this toy example, external input to neuron *e* or *f* results in periodic spike patterns with different frequencies, as shown in Fig. 3 (iii,iii'). Notice that the sets of neurons involved in each group overlap: neurons *a* and *b* are involved in both groups, and spikes of both *a* and *b* are required to depolarize *c* and *d*, but due to differences in conduction delays, the specific temporal pattern of *a* and *b* spikes results in phase and/or frequency differences in the cyclic pattern. Simulations in Izhikevich (2006) showed that many polychronous groups can emerge from a population of excitatory and inhibitory neurons whose synaptic weights evolve according to a spike-timing dependent plasticity learning rule (i.e., Hebbian learning), starting from an initial condition with random weights, connectivity, and delays.

Polychronous groups are a viable neural mechanism for gestural excitation for several reasons. First, the neurons that comprise a particular polychronous group are very often members of multiple such groups, so there is no need to view distinct groups as spatially distinct. Second, different groups within a population can be associated with different periods/frequencies of pattern repetition. Consequently, we can speculate that excitable modes of gesturals are associated with different but overlapping subsets of polychronous groups within the population—not unlike a bank of oscillators. As for why the neurons of a gestural population would exhibit this sort of internal organization, a plausible hypothesis is that longer-timescale dynamics of learning adapt gesturals to interact with another type of system: coordination modes.

Coordination modes. Another novel premise of mode-coupling theory is that there is a domain-general but nonetheless highly constrained system for the coordination and sequencing of actions. This system is distinct from gesturals, and it is instantiated in cortical-subcortical circuits—most likely cortico-basal ganglia-thalamic loops and the (pre-)supplementary motor area (Guenther & Hickok, 2016; Liu et al., 2025). This system is hypothesized to consist of pairs of sub-systems (coordination modes, or μ-systems) that are distinguished on the basis of their *sign* and *level*. The level distinction of μ-systems is presumed to map to a discrete set of frequencies in the beta-band range of oscillation (approximately 13-30 Hz), which has been implicated in the preparation of motor control (Engel & Fries, 2010; Kilavik et al., 2013). By convention the "highest" level is the one with the highest frequency and is depicted in the highest vertical position in schemas, but this level is given the lowest index. The sign distinction indicates whether a μ-system is coupled in-phase (+) or out of phase (-) to the oscillation associated with its level. $\{\mu^+, \mu^-\}$ pairs are viewed as an emergent behavior of cortical-subcortical circuits, rather than being neural populations localized to a particular area of the central nervous system. The phenomenological motivation for coordination modes is that they provide a mechanism for actions to be flexibly coordinated and/or sequenced without requiring direct interactions between the systems that govern action.

Fig. 4 shows how coupling of gestural excitations to coordination modes allows for flexibility in coordination and sequencing. Think of a word form (e.g. *poppy*) as a sequence of three groups of articulatory actions, {Pa}{P}{i}, where "P" refers to distinct excitations of a bilabial closure gestural ([LA clo]) and [a] and [i] refer to excitations of vocalic gesturals. Note that bilabial closure is associated with two separate groups. The word meaning itself is associated with an external system (a

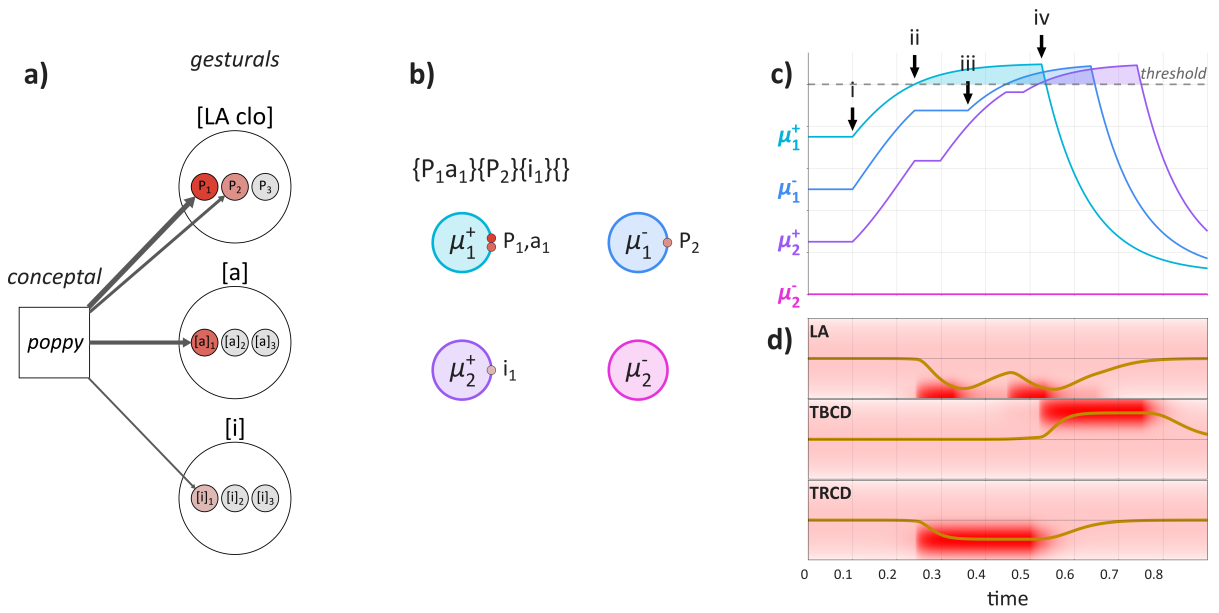


Figure 4: Flexibility of action sequencing through coupling of gestural excitations and coordination modes. (a) The conceptal associated with *poppy* activates gestural excitations (distinguished by numbered subscripts); node shading and arrow thickness indicate degree of activation. (b) Excitations couple to coordination modes (μ_1^+, μ_1^-, μ_2^+, μ_2^-). (c) Competitive selection: activations of the coordination modes over time, with shading indicating the period in which each mode exceeds a selection threshold. Arrows indicate (i) the onset of selection dynamics, (ii) the first selection, (iii) degating of the second mode, and (iv) suppression of the first mode. (d) Control field time series for the lip aperture (LA), tongue body constriction degree (TBCD), and tongue root constriction degree (TRCD), with field centroids (solid curves).

conceptal, see below) that activates the internal excitations of gesturals to varying degrees, as in Fig. 4 (a). The mode-coupling model described below couples gestural excitations to coordination modes (μ_m), the result of which is termed a *configuration*, shown in Fig. 4 (b). Typically more highly active gestural excitations will couple to higher-level coordination modes, but there are a variety of additional constraints the formation of such configurations.

Competitive selection and feedback. Sequencing is accomplished via a mechanism of *competitive selection* (Tilsen, 2013), which is a modification of *competitive queuing* (Bohland et al., 2010; Grossberg, 1978, 1987; Houghton & Hartley, 1995). There is a variety of behavioral evidence for this mechanism (Dell, 1986; Rhodes et al., 2004; Sternberg et al., 1988; Sternberg et al., 1978) and some neural evidence (Averbeck et al., 2002; Kornysheva et al., 2019; Silver et al., 2012). Competitive selection dynamics are shown in Fig. 4 (c). In competitive selection, feedback and gating systems result in a series of partially overlapping temporal epochs in which heightened activation is given to a particular coordination mode, and thereby to the gestural excitations that are coupled to it. Starting from an initial pattern of activation that corresponds to μ-system levels and signs (i), the most highly active μ-system exceeds the selection threshold first (ii) and its corresponding gestural excitation modes are

given heightened activation with phase-dependent constraints. Feedback systems induce degating of the next mode (iii) and suppression of the initially selected μ-system (iv). Selection and suppression cycles recur until all such systems have been selected and suppressed. The timeseries of vocal tract control fields in this example is shown in Fig. 4 (d). Importantly, in competitive selection, action sequencing derives entirely from the application of change rules to system states. The conceptual advantage of this is that regularity in the temporal order of action arises indirectly from the evolution of activation states, rather than being information that is conceptualized as temporal order *per se*.

There are important consequences to whether the feedback systems involved in suppression and gating are external (involving peripheral sensorimotor systems and basal ganglia-cortico-thalamic circuits) or internal (involving predictive forward models most likely instantiated with cortico-cerebellar circuits; Tanaka et al., 2020; Wolpert et al., 1998). The crucial difference is that internal feedback does not interact with the surroundings (on utterance timescales) and thus is well-suited for generating predictive feedback states. This in turn allows suppression and degating to be partly dissociated. Looking at a variety of empirical patterns, Tilsen (2016) argued that in early speech development, children control action sequencing primarily with external feedback, and as they acquire more reliable predictive feedback models, competitive selection comes to rely more on internal feedback systems. Theoretical and experimental work supports the hypothesis that speakers can intentionally control speech rate by adjusting their reliance on internal vs. external feedback (Tilsen, 2022, 2024), with internal feedback reliance resulting in faster speech and greater overlap of articulatory actions.

Conceptal activation. The concept-related systems referenced above are typically associated with meanings or ad-hoc action intentions—and hence referred to as *conceptals*. Conceptals are presumed to be distinguishable on the basis of their interactions with sensory systems, motor systems, and other conceptals. This paper will not discuss the dynamics of conceptals in any detail, but it is supposed that, like gesturals, they are neural populations predisposed to collective oscillation, likely distributed in prefrontal and multimodal association areas, and that what we experience as meaning corresponds to transiently stationary periodic state trajectories of conceptals (see Tilsen, 2019c). Here we will assume that conceptals, like coordination modes, are selected in the course of generating speech, and so doing, they activate gestural excitation modes. The patterns of gestural activation associated with conceptals are encoded in long-term memories of word forms/concepts; microscopically these are factors contributing to synaptic efficacy of projections from neurons in conceptal populations to neurons in gestural populations.

2 Model implementation

The model discussed here is comprised of three primary mechanisms: (i) mode coupling, (ii) mode selection, and (iii) vocal tract state control via field activation. Vocal tract state control has been described elsewhere (Kim & Tilsen, 2025; Tilsen, 2018, 2019a) and is an extension of the standard Task Dynamics model (Saltzman & Munhall, 1989). Mode selection is a revision of competitive queuing (see Section 1.3;

Bullock, 2004; Grossberg, 1987; Tilsen, 2013, 2022) that incorporates internal and external feedback to govern the gating and suppression of coordination modes. Because vocal tract state control and mode selection are not the primary focus of this paper, detailed descriptions of these are provided in supplementary material. The mode-coupling mechanism presented below is a novel contribution, refining and elaborating on ideas described in Tilsen (2019c).

2.1 Mode coupling

Assume a set C of conceptals. When a speaker produces or subvocally simulates speech, a subset of these become simultaneously active above a threshold (i.e., *selected*). A conceptal selection epoch is a period of time in which some conceptal(s) are above a selection threshold; during such periods the influences of the selected conceptals on gesturals are assumed to remain constant.

Assume a set G of gesturals g_i with $i \in 1, \ldots, n_g$. Each gestural is comprised of distinct subsystems—internal excitation modes. We index all gestural excitations (g-systems) with a single index, writing e_j for excitation j, and define $g(j) \in G$ as its parent gestural system. Each gestural excitation is associated with an $[r, \theta, f]$ (amplitude, phase, frequency) tuplet.

Assume a set M of coordination modes μ_m (or, μ-systems). A coordination mode can also be identified by its level and sign as μ^s_ℓ, with $\ell \in 1, \ldots, n_\ell$ and $s \in \{+, -\}$; thus m identifies an individual mode, whereas ℓ identifies a mode pair. $s_m \in \{+1, -1\}$ is the sign of mode m.

In a given conceptal selection epoch, selected conceptals increase the amplitudes of gestural excitations they are associated with. We refer to this as conceptal-gestural *activation* and assume that activation values are determined by long-term memory/learning mechanisms. In other words, the component of lexical memory that corresponds to phonological form is a vector of scalar values in the space of gestural excitation amplitudes.

Gestural excitation states subsequently evolve to minimize a free energy, subject to constraints on their interactions with μ-systems and other g-systems. The coupling of a gestural excitation to a coordination mode is assumed to involve a rapid phase- and frequency-locking of the g-system to a μ-system. Rather than modeling the fast temporal dynamics of coupling, we adopt a discrete, coarse-grained description in which the system state—a mode-coupling configuration (or, g-μ configuration)—encodes (i) which g-systems are coupled to which μ-systems, (ii) the phases of g-systems (relative to the μ-system to which they are coupled), and (iii) the amplitudes of g- and μ-systems. Free energy is defined as a function of this state. Section 4 elaborates whether this energy can be interpreted as physical.

In order for the evolution of the system to result in a configuration that is phenomenologically compatible with empirical observations of articulatory patterns, conceptal-gestural activation must be located in a subregion of the gestural amplitude subspace in which relative activations of gestural excitations predispose them to interact with μ-systems in an appropriate way. In effect, conceptal-gestural activation imposes additional constraints on g-μ state trajectories, which ultimately constrains how actions are temporalized (via mode selection and vocal tract control). Crucially, conceptal-gestural activation is not *per se* a specification of temporal

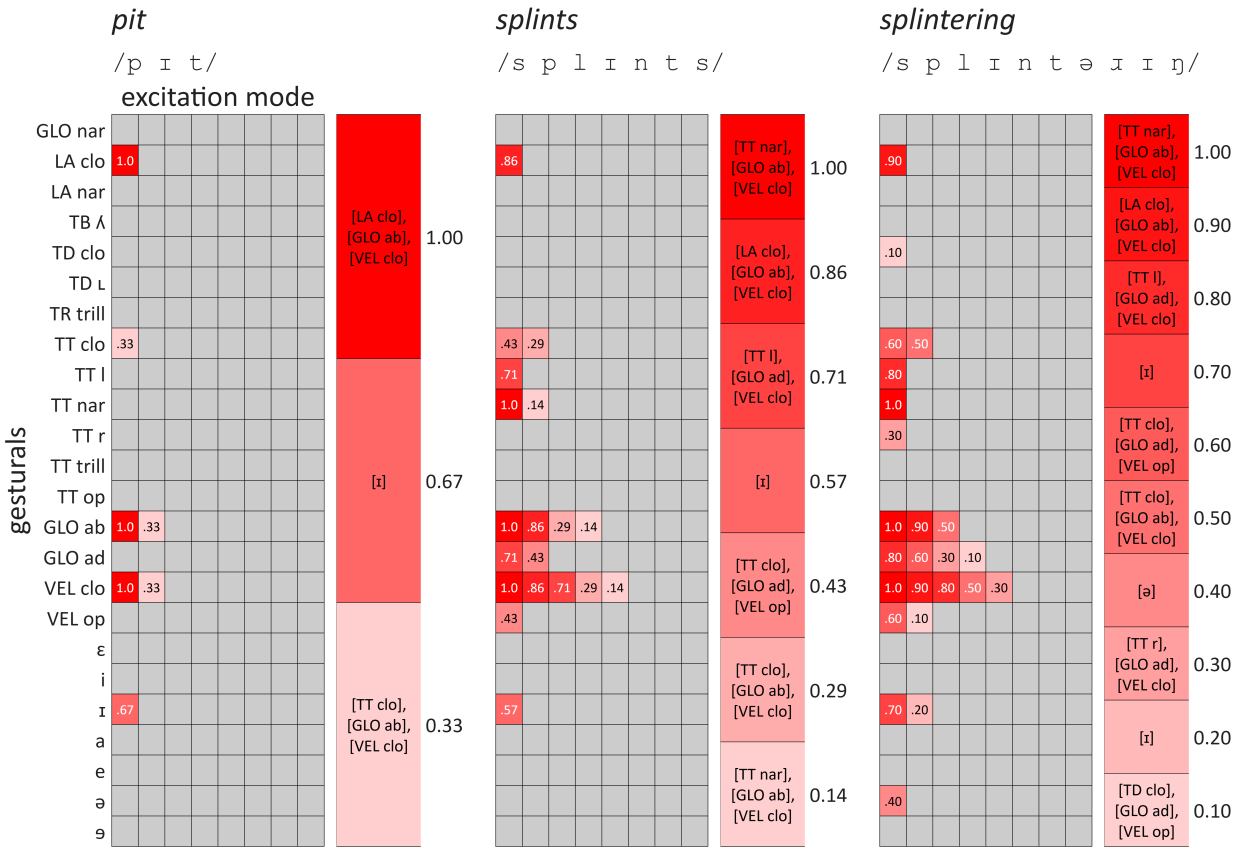


Figure 5: Examples of conceptal-gestural activation for word forms *pit*, *splints*, and *splintering*. Rows are gesturals and columns are excitation modes. For clarity only a subset of gesturals are shown.

order; instead, it represents the amplitudes of internal excitation modes of gesturals when a conceptal is selected.

Here a fairly simple algorithm was used to determine conceptal-gestural activation, based on the gestural composition and timing patterns expected from a conventional linear phone/segmental analysis of the form associated with a conceptal. A phone-gesture map was constructed for this purpose. Specifically, the algorithm works as follows: (i) set the phone sequence index $p = 1$, and initialize an $n_g \times n_{\text{e-modes}}$ array $\mathcal{E}$ of zeros with rows corresponding to gesturals and columns to gestural excitation modes, where $n_{\text{e-modes}}$ is the number of internal excitation modes available for gesturals. (ii) For each gestural associated with phone p, assign the value $1 - \frac{p-1}{n_p}$ to the first zero column of the array in the row corresponding to that gestural. (iii) Increment p. Repeat steps (ii) and (iii) until $p > n_p$, where n_p is the number of phones in the linear segmental analysis. Note that for complex segments (e.g. affricates), the narrowing gestural is assigned a value halfway between that of phone p and phone $p + 1$, so that it is less active than the closure gestural. Some examples of conceptal-gestural activation patterns generated by this algorithm are shown in Fig. 5.

It is critical to emphasize that mode-coupling theory does not presuppose that the nervous system implements the above algorithm; the algorithm is merely a heuristic.



Figure 6: Coarse-grained state space trajectory in which gestural excitations of the word pit couple to a pair of coordination modes. In each panel, uncoupled gesturals are shown along the top, and μ-coupled gesturals are located on circles that represent coordination modes.

Conceptal-gestural activation must be learned and refined over the course of development, via microscale synaptic modulation/growth processes. Thus even though the above algorithm makes use of a linear ordering of segments for convenience, the theory makes no presupposition that a linear order of segments be encoded in memory. Rather, what conceptal memories encode is merely a pattern of activation of gestural excitations. It seems likely that these patterns could be learned and encoded in the weights of a multi-layer neural network, but this has yet to be verified.

Given conceptal-gestural activation values, mode-coupling couples gestural excitations to coordination modes. To show how this works, we first consider a quasi-deterministic dynamics for the coupling process, in which the next state is always the state that represents the largest reduction in energy relative to the current state.

An example of a temporal sequence of configurations for the conceptal [PIT] (/pɪt/) is shown in Fig. 6. The circles in each panel are coordination modes, i.e. μ-systems. Uncoupled g-systems are shown at the top of the panel, shaded according to their conceptal activation. g-systems that are coupled to a μ-system are depicted as points on the circles. The occupied point on the circle indicates φ_j, the phase of the g-system relative to the phase of the μ-system to which it is coupled. Each panel is labeled with a textual representation of the state, where gestural excitations within curly brackets are μ-coupled, and those outside of curly brackets are uncoupled. Bracket pairs thus correspond to μ-systems and are always presented in order of mode pair level and sign, i.e. $\mu_1^+, \mu_1^-, \mu_2^+, \mu_2^-, \ldots$

The following conventions are adopted for the representation of gestural excitations: vocalic gestural excitations are represented by international phonetic alphabet (IPA) symbols (e.g. ɪ, æ, etc.); oral consonantal gestural excitations by capitalized IPA symbols for voiceless consonants (e.g. P = labial closure, T = alveolar closure, K = velar closure, etc.); velum opening/closing excitations by empty/slashed squares (□/⧅); glottal abduction/adduction excitations by empty/barred circles (○/⦶); laryngeal rais-

ing/lowering excitations by vertical arrows (↑/↓); and labial protrusion/retraction by horizontal arrows (←/→).

To simplify the depiction of configurations, when all of the gesturals associated with a phone are coupled to the same mode with the same relative phase (which we refer to as an *orbit*), rather than depicting all three gesturals in the orbit, we collapse them into the appropriate IPA consonant symbol, enclosed in [] brackets. For example, in the configuration obtained in the fourth iteration of Fig. 6, P (bilabial closure), ○ (open glottis), and ⦸ (closed velum) excitations are coupled to same orbit (here the 0-phase of μ_1^+); these are replaced with the symbol [p] (a voiceless bilabial oral stop). It is important to emphasize that the collapsed symbols are used solely for visual convenience, and this convention does not imply that phones are cognitive entities.

In the deterministic model of a system state trajectory, there is one configurational change in each iteration, and that change is always the one associated with the largest reduction in energy. The allowable changes are one of two kinds: (i) *coupling*, in which a previously uncoupled gestural excitation couples to any one of the coordination modes; or (ii) *demotion*, in which all of the excitations that occupy one orbit of a mode (i.e. that share its μ-system and relative phase) move together to the next mode in the sequence $\mu_1^+, \mu_1^-, \mu_2^+, \mu_2^-, \ldots$, so that an orbit of μ_ℓ^+ moves to μ_ℓ^- and an orbit of μ_ℓ^- moves to $\mu_{\ell+1}^+$. Excitations coupled to the lowest mode cannot be demoted.

Fig. 7 illustrates how configuration changes are determined from one iteration to the next. The left side of the figure shows the current configuration at iteration 5 and the configuration that is selected in this iteration. In this case, excitations associated with [p] (P, ○, ⦸) and [ɪ] have already been coupled to μ_1^+, but excitations associated with [t] (T, ○, ⦸) have not yet coupled to a μ-system. The topmost array on the right shows a break-down of contributions to the energy current configuration, with total energy shown outside the lower right corner of the array. The remaining arrays show energy contributions for alternative configurations that are accessible with a single configurational change at this iteration. Note that one of the alternatives shown is equivalent to the current state, representing no configuration change. In general, at any given iteration of the mode-coupling process, there is a moderately sized set of accessible configurations, depending on how many gestural excitations are present and how many μ-systems are active. As gestural excitations couple to μ-systems, fewer alternative configurations become accessible.

The deterministic model simply transitions at each iteration to the lowest-energy configuration that is available. At iteration 5, this lowest energy configuration corresponds to the outlined array in Fig. 7, a configuration in which an internal excitation of the alveolar closure gestural (T) couples to μ_1^-. Because this is the lowest-energy available configuration, iteration 6 will begin with (T) coupled to this mode. Thus it is the contributions to configuration energy that are responsible for determining how gestural excitations are organized into coupling relations with μ-systems. Each contribution corresponds to a column of the component energy arrays; these are sorted according to weighting terms (w) in order of decreasing weight, with values shown in Fig. 7. Each row corresponds to a particular μ-system, sorted by level and sign, i.e. $\mu_1^+, \mu_1^-, \mu_2^+, \mu_2^-$,... Each column can be viewed as a term in a function that calculates the total energy of the system state.

Gestural properties. Fig. 8 shows the classification of gesturals adopted in the model.

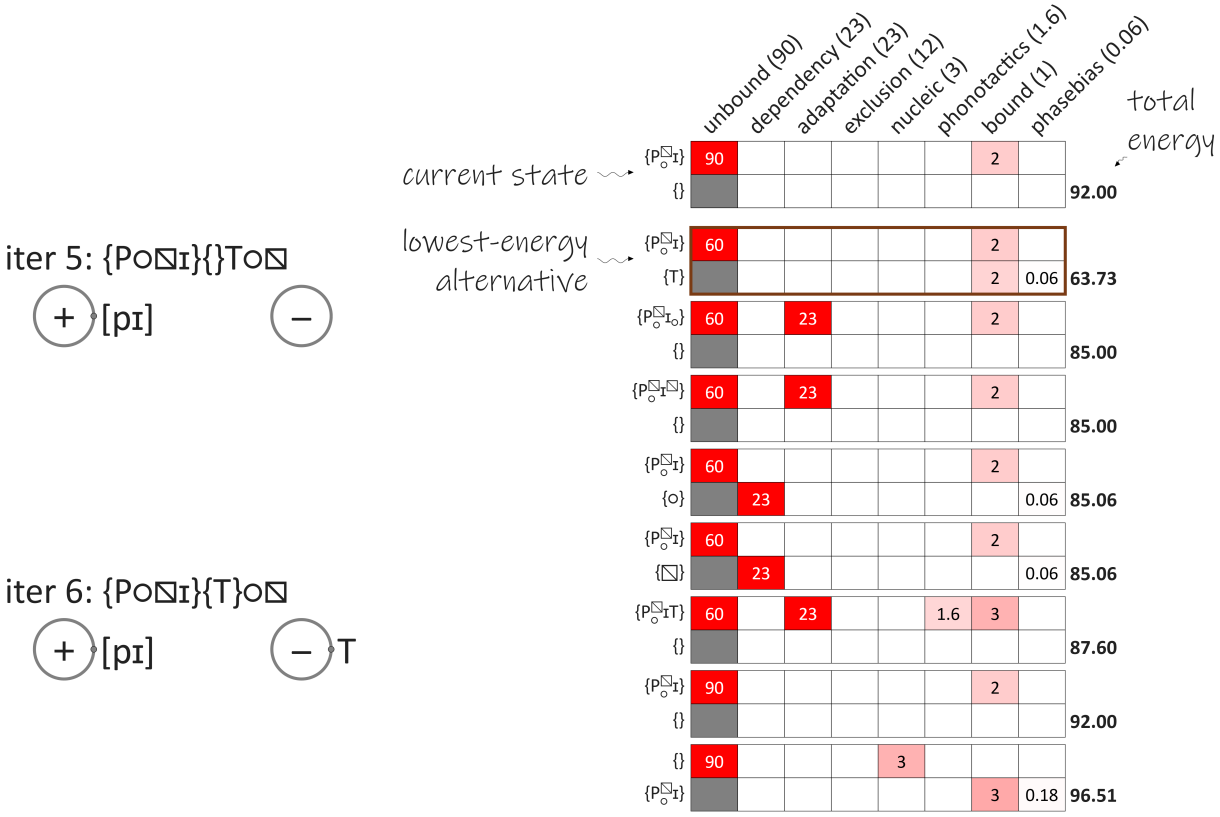


Figure 7: Example of one iteration of state change in mode coupling. Left: current and next configuration. Right panels: current energy details (top-most panel) and accessible configuration energy details (all but top-most panel). Total energy is shown to the right of each panel. Within each panel, rows correspond to μ-systems. Gestural excitations coupled to a mode are shown in row headers.

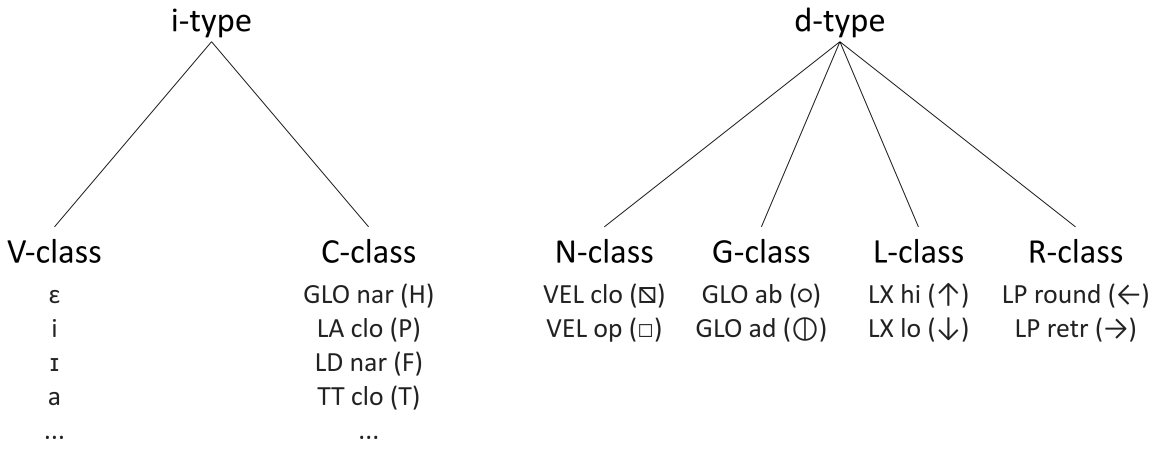


Figure 8: Classification of gesturals. All gesturals (and their internal excitations) are i-type (independent) or d-type (dependent). Within i- and d-types several sub-classes are defined.

All gesturals (and their internal excitations) are hypothesized to belong to one of two types, independent (i-type) or dependent (d-type). Within i-type gesturals, there are two classes: V-class (vocalic; involving lingual articulatory gestures associated with vowel postures), and C-class (consonantal; involving lingual, labial, or glottal consonantal constriction gestures). Within d-type gesturals, there are four classes: N-class (nasal; associated with opening/closing the velum), G-class (glottal voicing; associated with abducting/adducting the vocal folds), L-class (laryngeal elevation; associated with raising/lowering the larynx, which occurs in ejective/implosive sounds), and R-class (labial rounding/retraction). Note that [GLO nar] and [GLO clo] are C-class, being associated with [h] and [ʔ], respectively, while [GLO ab] and [GLO ad] are G-class.

Types and classes play a large role in the calculation of configuration energy. Type/class hypotheses are ultimately phenomenologically motivated: we adopt them because doing so facilitates an empirically adequate characterization of phonological patterns. Nonetheless, there are independent rationales for the classification. Regarding gestural types, the distinction between i-type (independent) and d-type (dependent) gesturals is motivated by the observation that the articulatory actions associated with d-type gesturals never (or perhaps, rarely) occur in the absence of actions associated with i-type gesturals. In energy calculations (detailed below), this distinction is manifested as a cost for coupling d-type gestural excitations to μ-system orbits without a corresponding i-type gestural excitation. All V- and C-class gesturals are i-type, whereas all N/G/L/R-class gesturals are d-type.

Regarding gestural classes, vowel sounds (products of V-class gesturals) always involve a relatively unconstricted lingual posture. This is not the case for consonants. Hence there are somatotopic/functional differences between V-class and C-class gesturals, manifested in their influences on motoric and somatosensory fields. Within d-type gesturals, sub-classes of N (nasal), G (glottal voicing), L (laryngeal elevation), and R (lip protrusion/retraction) are also somatotopically motivated: the sensorimo-

tor neural fields that these classes of gesturals project to differ.

In general, V-class gesturals are more constrained in their abilities to couple to μ-systems than C-class gesturals: each $\{+\}$ mode should contain exactly one nucleic V-class excitation (nucleus occupation), V-class activations set the activation floor of their mode (frequency adaptation), and their preferences for coupling to $\{+\}$ or $\{-\}$ modes are encoded in long term memory. This latter property is implemented as sign property of V-class gesturals, and in effect it holds that each vocalic gesture may be associated with two gesturals, one that is nucleic (preferring to couple to $\{+\}$ modes), and one that is non-nucleic (preferring to couple to $\{-\}$ modes). The differences between V- and C- class gesturals can be interpreted as a difference in the asymmetry of interactions between these gesturals and coordination modes: vocalic gesturals are likely to have stronger influences on μ-systems than consonantal ones do. In other words, interactions with μ-systems are less asymmetric for V-class than C-class gesturals.

Coordination modes and gestural phasing. Coordination modes (μ-systems) are distinguished according to their sign (i.e. μ_ℓ^+ vs. μ_ℓ^-) and their level ($\ell = 1, \ldots, n_\ell$). For convenience we refer to {+} and {−} modes when level is not relevant. The sign distinction has a number of consequences for configuration energy. In general, exclusion effects obtain only between gestural excitations that occupy the same orbit (i.e., ones that are coupled to the same μ-system with the same relative phase). For all coupled g-systems, relative phase is defined as $\varphi_j = \theta_j - \theta^\mu_{m(j)}$, where θ_j is the absolute phase of excitation e_j, θ^μ_m is the absolute phase of coordination mode μ_m, and $m(j)$ indexes the mode to which excitation e_j is coupled. The phase difference between excitations is $\Delta\theta_{jk} = \theta_j - \theta_k$. The φ values of all the gestural excitations coupled to a particular coordination mode are derived from solutions to a system of phase-coupled oscillators, and these values are updated at each iteration of the mode-coupling process. The numerical phase solver treats each coordination mode and its coupled excitations as one oscillator network. Let $\mathcal{O}_m$ contain those excitations and the mode oscillator itself, and let u, v index members of this network. The phase update rule is

$$\begin{aligned} \dot{\theta}_u &= \omega - \kappa \sum_{v \in \mathcal{O}_m} \Phi_{uv} \sin(\theta_u - \theta_v), \\ \theta_u(t + \Delta t) &= \theta_u(t) + \Delta t\, \dot{\theta}_u(t). \end{aligned} \tag{1}$$

Here ω is a common natural angular frequency, κ is a scaling factor, and Δt is the numerical solution timestep. Eq. (1) is a Kuramoto-type model of coupled phase oscillators (Acebrón et al., 2005; Kuramoto, 1984), which differs from the classic form in that the oscillators share a common natural frequency, the coupling is not normalized by the number of oscillators, and the coupling matrix is signed, so that oscillators may attract or repel one another (Hong & Strogatz, 2011). The entry Φ_{uv} specifies the influence of oscillator v on oscillator u. Positive entries produce attraction toward zero phase difference, and negative entries produce repulsion toward π modulo 2π. In order to speed up simulations, the mode-coupling model does not implement the numerical phase solver on each iteration. Instead, μ-relative phase values were precomputed from steady-state solutions of Eq. (1), and numeric placeholders for these solutions are imputed to the relative phases of gestural excitations on each iteration:

i-type excitations receive the placeholder phase pattern of their mode, and each d-type excitation takes the phase of the i-type excitation with the same activation.

The excitation-to-excitation block of Φ (the portion of this matrix that encodes gestural excitation relative phase coupling forces) is K. For {+} modes, $K_{jk} = -1$ between distinct C-class excitations. For negative modes, $K_{jk} = -1$ between distinct i-type (C- or V-class) excitations. $K_{jk} = 1$ when j is d-type, k is i-type, and the two have equal activation (i.e. are associated with the same phone in the activation heuristic), reflecting the dependency of d-type excitations on i-type ones. All other excitation-to-excitation entries are zero, including direct C-class–to–V-class coupling in positive modes. The μ-coupling entries in Φ are $\Phi_{j\mu_m} = \Phi_{\mu_m j} = 1$ for all i-type gesturals.

From the above constraints, a small set of stable φ-configurations is available, shown in Fig. 9. Panels (i)–(i″) show φ-configurations for {+} modes. When a single C-class gestural excitation (here [LA clo]) is coupled to the mode, along with a V-class one ([a]), the two couple to the same 0-phase orbit (Fig. 9 (i)), along with the d-type gesturals (here [GLO ab] and [VEL clo]). When multiple C-class gesturals (i.e. 2 or 3) are coupled to the {+} mode, the repulsive forces between them result in non-zero phase offsets. Panels (ii)–(ii″) show available configurations for {−} modes, where all i-type gesturals are repulsively phase coupled. A single i-type gestural (ii) couples to {−} with 0 phase offset, but multiple i-type gesturals have phase offsets. The exact values of the phase offsets of gesturals depend on magnitudes of non-zero elements in Φ, and in particular the relative strengths of the repulsive forces between gesturals and attractive forces with μ. For any two gesturals coupled to the same μ-system, the maximal absolute relative phase difference is π.

It is important to note that the above system does not describe the *process* of g-μ coupling, only the phase dynamics of g-systems and a μ-system. Qualitatively, μ-systems pull i-type g-systems toward phase 0, but repulsive forces that obtain between i-type g-systems prevent those phases from collapsing to 0. In contrast, d-type g-systems simply match the phase of an associated i-type. The solutions of the relative phase dynamics provide an effective ordering of i-type gesturals according to their phases, and for this solution to match empirical patterns, it is assumed that the initial phase differences conform to inequality relations in conceptal-gestural activation. This assumption ensures that timing relations manifested through mode selection/vocal tract control are empirically consistent. The inequalities can be expressed as phase precedence relations. For example, for a {+} mode representing a $C_1C_2C_3V$ form, $\theta^0_{C1} > \theta^0_{C2} > \theta^0_{C3}$, with $-\pi/2 < \Delta\theta_{jk} < \pi/2$.

Metrical activation. Metrical stress patterns arise from a state variable (*metrical activation*) associated with coupling mode *pairs*, which we refer to as σ-systems (syllable systems) and index with level ℓ of the mode pair. On each iteration of mode-coupling, metrical activation values are calculated dynamically from interactions within a network of σ-systems subject to external inputs. Metrical activation dynamics are iterated until a stable activation pattern arises across σ-systems, reflecting a balance between inputs and interactions between systems on adjacent levels.

The governing equation for σ-system activation is shown below, where the highest level is the level with the highest mode frequency, i.e. the lowest level index, $\ell = 1$. The total input (I_ℓ) to σ_ℓ is a sum of input to the highest level syllable (I^b, begin-input), input to the lowest level syllable (I^e, end-input), input based on syllable weight (I^{μ^-}, deriving from occupation of the {−} mode), and accentual input ($I^{\mathcal{L}}$). The total

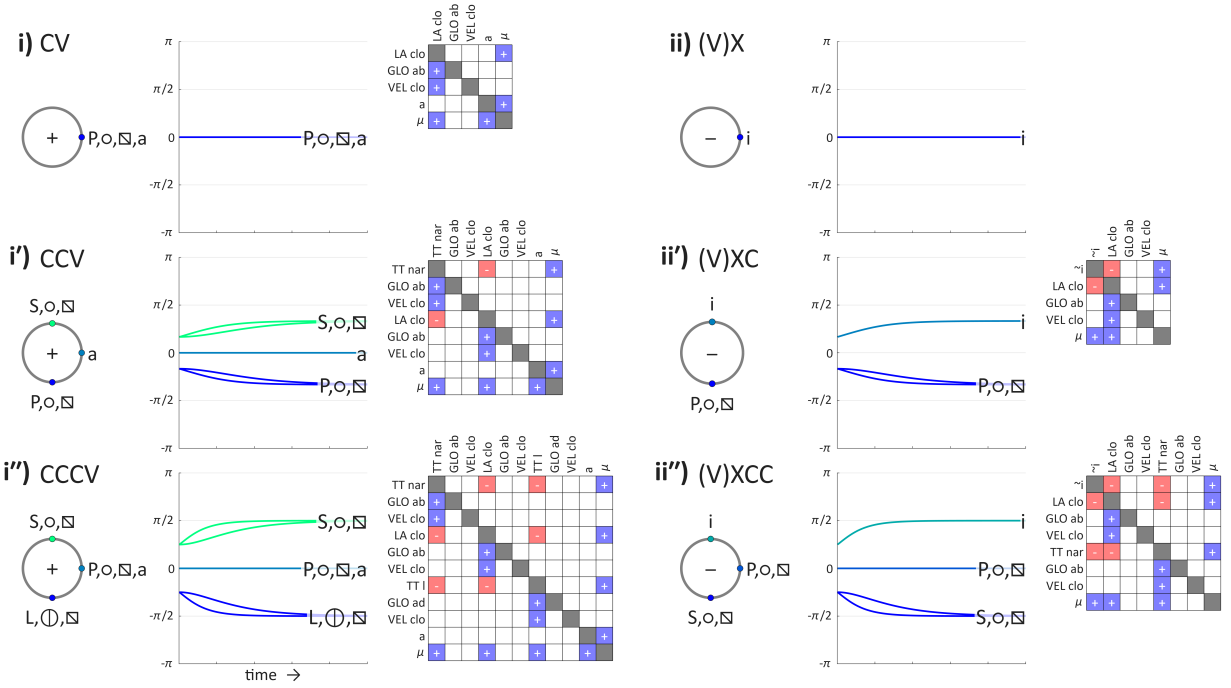


Figure 9: Coupling matrices, relative phase dynamics, and relative phase configurations. (i)–(i″): {+} modes with 1, 2, and 3 C-class gesturals, representing CV, CCV, and CCCV forms, respectively. (ii)–(ii″): {−} modes with 1, 2, or 3 i-type gesturals, representing VX, VXC, and VXCC forms, respectively.

is nonzero only for syllable systems whose {+} mode is occupied. σ-systems exert inhibitory forces on the σ-systems of levels immediately above and below (i.e. on $\sigma_{\ell-1}$ and $\sigma_{\ell+1}$). In the absence of accentual and internal weight inputs, an input at $\ell = 1$ or $\ell = n_\ell$ (the highest or lowest level σ) gives rise to an alternating pattern of stress, because neighboring-level interactions are inhibitory. In other words, the periodicity of default stress patterns arises from a lateral-inhibition dynamics (see Goldsmith (1994) for a similar model, Tilsen (2019b) for a standing wave variation, and Iskarous and Goldstein (2018) for a related alternative).

$$I_\ell = [\![\mu_\ell^+ \text{ occupied}]\!] \left(I_\ell^{\mu^-} + I_\ell^{\mathcal{L}} + I^b [\![\ell = 1]\!] + I^e [\![\ell = n_\ell]\!] \right). \tag{2a}$$

In Eq. (2a) the indicator $[\![P]\!]$ takes the value 1 when the enclosed condition is true and 0 otherwise. Each input term comes from a raw input scaled by a weight, so the weights do not appear separately in the equation. Begin- and end-inputs have default weight 1; for English $I^b = 0$ and $I^e = -1$. The internal input $I_\ell^{\mu^-}$ is nonzero when μ_ℓ^- is occupied. Lexical input $I_\ell^{\mathcal{L}}$ represents accentuation of nucleic V-class excitations coupled to σ_ℓ. Note that accentuation is a property solely of nucleic V-class excitations, which allows for long-term memory (lexical) specification of stress/accent. Activation is initialized at $\sigma_\ell^{(1)} = I_\ell$ and updated as follows:

$$\sigma_\ell^{(k+1)} = \sigma_\ell^{(k)} + \alpha \tanh\left(-\sigma_{\ell-1}^{(k)} - \sigma_{\ell+1}^{(k)} \right) + \lambda \left(I_\ell - \sigma_\ell^{(k)} \right) - \gamma \sigma_\ell^{(k)}. \tag{2b}$$

The update combines interaction with neighboring levels, attraction toward the input, and linear decay. The parameter α weights the neighboring-level interaction, λ weights the total input, and γ is a decay rate; their default values are $\alpha = 2/3$, $\lambda = 1$, and $\gamma = 1/10$. The boundary condition $\sigma_0^{(k)} = \sigma_{n_\ell+1}^{(k)} = 0$ ensures that systems at the highest and lowest levels have only one neighbor. Iteration continues until the change in the activation pattern is below a tolerance, and the final activation is normalized as:

$$\bar{\sigma}_\ell = \frac{\left[\sigma_\ell^{(k^*)} \right]_+}{\max_{\ell'} \left[\sigma_{\ell'}^{(k^*)} \right]_+}. \tag{2c}$$

Note that k^* denotes the final iteration and $[\,\cdot\,]_+$ is half-wave rectification, so that σ with negative activation are treated as unstressed and the remaining positive activations $\bar{\sigma}_\ell$ fall in $[0, 1]$. For calculation of configuration energies, both μ_ℓ^+ and μ_ℓ^- associated σ_ℓ are assigned the activation $\bar{\sigma}_\ell$.

Total energy. The energy of the current state of the system has two main components: the energy associated with unbound gestural excitations, E_{unb}, and the bound configuration energy, E_{bound}. The former comes from gestural excitations that are not coupled to coordination modes; the latter from the configuration of excitations that are coupled to coordination modes:

$$E_{\text{total}} = E_{\text{unb}} + E_{\text{bound}} \tag{3}$$

Let j and k index gestural excitations, let m index coordination modes, and write M^+ and M^- for the sets of {+} and {−} modes. Define the coupling matrix χ, with rows corresponding to modes and columns corresponding to gestural excitations:

$$\chi_{mj} = \begin{cases} 1, & \text{if excitation } j \text{ is coupled to mode } m, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Let $\mathrm{cl}(j) \in \{V, C, N, G, L, R\}$ denote the class of excitation j. $V_j = [\![\mathrm{cl}(j) = V]\!]$, $C_j = \ldots$, etc. are class indicators. $\mathcal{I}_j = [\![\mathrm{cl}(j) \in \{V, C\}]\!]$ and $\mathcal{D}_j = [\![\mathrm{cl}(j) \in \{N, G, L, R\}]\!]$ are type indicators. V_j^+ and V_j^- identify V-class excitations of gesturals with positive and negative sign—a property of the gestural itself, not of the mode it couples to. a_j denotes the activation of excitation j. $\mathcal{M}_j = \sum_m \chi_{mj} = 1$ when excitation j is coupled to a coordination mode, 0 otherwise (each excitation couples to at most one mode). For coupled excitations, $m(j)$ denotes the index of the mode to which excitation e_j is coupled, so that this mode is $\mu_{m(j)}$, and φ_j is the phase of the excitation relative to the mode. The mode sign of a coupled excitation is $s_j \equiv s_{m(j)}$. With the above conventions, the q-state (quantized state) of a coupled excitation j is a tuplet $(\mu_{m(j)}, \varphi_j, \mathrm{cl}(j))$, and we define a q-state identity relation:

$$Q_{jk} = [\![j \neq k]\!][\![q_j = q_k]\!] \tag{5}$$

Unbound excitation energy. The unbound excitation energy E_{unb} for an uncoupled gestural excitation is proportional to its activation. This favors the coupling of more strongly activated gestural excitations before less strongly activated ones. Differences in gestural excitation activations are exogenously determined by conceptal systems, and hence conceptals constrain the temporal order in which gestural excitations couple to coordination modes.

$$E_{\text{unb}} = w_{\text{unb}} \sum_j a_j \left(1 - \mathcal{M}_j\right). \tag{6}$$

The bound configuration energy E_{bound} is decomposed into separately weighted contributions:

$$\begin{aligned} E_{\text{bound}} = {} & \underbrace{w_{\text{nuc}} E_{\text{nuc}}}_{\substack{\text{nucleus} \\ \text{occupation}}} + \underbrace{w_{\text{excl}} E_{\text{excl}}}_{\text{exclusion}} + \underbrace{w_{\text{dep}} E_{\text{dep}}}_{\text{dependency}} \\ & + \underbrace{w_{\text{phase}} E_{\text{phase}}}_{\substack{\text{phase} \\ \text{bias}}} + \underbrace{w_{\text{adapt}} E_{\text{adapt}}}_{\substack{\text{frequency} \\ \text{adaptation}}} + \underbrace{w_{\text{phon}} E_{\text{phon}}}_{\text{phonotactics}} + \underbrace{w_{\text{bind}} E_{\text{bind}}}_{\substack{\text{binding} \\ \text{energy}}} \end{aligned} \tag{7}$$

Binding energy (E_{bind}). Each coupled i-type gestural excitation incurs an energy cost that depends on the level and sign of the mode to which it is coupled and on the metrical activation of the corresponding σ-system. Let $\ell(m)$ denote the level of mode m, so that $\bar{\sigma}_{\ell(m)}$ is the (normalized) metrical activation of the syllable system $\sigma_{\ell(m)}$ to which m belongs. The binding energy associated with mode m is

$$\epsilon_m = \ell(m) + c_- [\![s_m < 0]\!] - c_{\text{metr}} \bar{\sigma}_{\ell(m)}, \tag{8}$$

where c_- is the additional energy associated with coupling to a negative mode and c_{metr} determines a reduction in energy due to metrical activation. In the current

implementation, the default values are $c_- = 2/3$ and $c_{\text{metr}} = 1/2$. The binding-energy contribution of mode m is the sum of the binding energies of all of its coupled i-type excitations; d-type excitations do not contribute. The total binding energy is therefore

$$E_{\text{bind}} = \sum_m \sum_j \epsilon_m \chi_{mj} \mathcal{I}_j. \tag{9}$$

Exclusion principle (E_{excl}). The exclusion term penalizes excitations that occupy the same q-state. Note that excitations are not subject to exclusion on the zero-phase orbits of {+} modes, and this is handled by the indicator $\zeta^+_{jk} = [\![s_j = s_k = +1]\!][\![\varphi_j = \varphi_k = 0]\!]$:

$$E_{\text{excl}} = \sum_m \sum_j \chi_{mj} \sum_k Q_{jk} \left(1 - \zeta^+_{jk}\right). \tag{10}$$

Dependency (E_{dep}). A d-type excitation incurs an energy cost when it occupies an orbit in which there is no i-type excitation. Two excitations are defined as occupying the same orbit when they are coupled to the same μ-system and have the same phase, which is expressed by the indicator $\Omega_{jk} = [\![\mu_{m(j)} = \mu_{m(k)}]\!][\![\varphi_j = \varphi_k]\!]$. Dependency energy is then:

$$E_{\text{dep}} = \sum_m \sum_j \chi_{mj} \mathcal{D}_j \left[\!\!\left[\sum_k \mathcal{I}_k \Omega_{jk} = 0 \right]\!\!\right] \tag{11}$$

A consequence of dependency, in combination with the exclusion principle, is that d-type excitations tend to couple to *partially filled orbits*. For a given d-type excitation, an orbit is partially filled when there is an i-type excitation but no d-type excitation of the same class as the given one.

Phase bias (E_{phase}). All excitations except V-class ones incur an additional energy cost when coupled to a {−} mode. This term implements a bias against the coupling of non-vocalic gestures to μ^- systems:

$$E_{\text{phase}} = \sum_m \sum_j \chi_{mj} \left(\neg V_j\right) [\![s_j = -1]\!]. \tag{12}$$

Frequency adaptation (E_{adapt}). There is an energy cost for coupled excitations whose activation is below an activation floor, which is determined by vocalic excitations associated with the relevant mode. For each mode m, the activation floor $a_m^{\min}$ is the minimum non-zero activation of the vocalic excitations coupled to that mode, with $a_m^{\min} = 0$ when no vocalic excitations are coupled. The effective floor $\tilde{a}_m^{\min} = a_m^{\min}$ for {+} modes, and $\tilde{a}_m^{\min} = a_{m+1}^{\min}$ for {−} modes:

$$E_{\text{adapt}} = \sum_m \sum_j \chi_{mj} [\![0 < a_j < \tilde{a}_m^{\min}]\!]. \tag{13}$$

Thus the frequency-adaptation term counts coupled excitations whose activation is insufficient relative to the activation floor associated with their mode. This reflects

an asymmetry in frequency adaptation: the frequency of a gestural excitation can decrease without additional energy to couple to a coordination mode, but an increase requires additional energy.

Nucleus occupation (E_{nuc}). {+} modes are stabilized by an appropriate configuration of V-class excitations. The energy cost is one unit for each {+} mode that does not contain exactly one V^+ excitation, and additional cost for each V^- excitation that is coupled to a {+} mode:

$$E_{\text{nuc}} = \sum_{m \in M^+} \left[[\![\sum_j \chi_{mj} V_j^+ \neq 1]\!] + \sum_j \chi_{mj} V_j^- \right]. \tag{14}$$

This term can be viewed as an instantiation of the phenomenological imperative that syllabic structures contain an appropriate nucleus: a positive mode is favored when it contains exactly one non-negative-sign vocalic excitation and disfavored for each negative-sign vocalic excitation.

Phonotactics (E_{phon}). Configurations can be constrained by language- or speaker-specific phonotactic interactions between gesturals. These constraints are represented as a set Π of prohibited ordered pairs of gesturals, which apply to all excitations of those gesturals. The ψ-state of an excitation is its q-state without the phase component, $\psi_j = (\mu_{m(j)}, \text{cl}(j))$. Two distinct excitations violate a prohibition when they have the same ψ-state and their gesturals are an ordered pair in the set Π. The ordering is taken to correspond to a phase inequality, so that a violation is $\Psi_{jk} = [\![\psi_j = \psi_k]\!][\![(g(j), g(k)) \in \Pi]\!][\![\varphi_j > \varphi_k]\!]$. Phonotactic energy is then:

$$E_{\text{phon}} = \sum_m \sum_j \chi_{mj} \sum_k \Psi_{jk}. \tag{15}$$

Energy weights. Each of the terms described above is weighted. Weights were initially determined through manual experimentation in the process of developing the model; these were subsequently used as seeds of an optimization procedure. The procedure fixed the binding energy weight at 1 and inferred values for other weights by requiring the model to accurately generate configurations expected for English word forms. More detail on the optimization procedure is provided in the online repository. The weights used in the current model are shown in Fig. 7, and are $w_{\text{unb}} = 90$, $w_{\text{adapt}} = 23$, $w_{\text{dep}} = 23$, $w_{\text{excl}} = 12$, $w_{\text{nuc}} = 3$, $w_{\text{phon}} = 1.6$, $w_{\text{bind}} = 1$, and $w_{\text{phase}} = 0.06$.

2.2 Stochastic mode coupling

The deterministic dynamics of mode configuration described above were obtained by selecting the most probable configuration at each iteration. These deterministic dynamics correspond to the limit where inverse temperature $\beta \to \infty$ (or temperature $T \to 0$). To model stochastic dynamics, a Boltzmann distribution is used. Let $x^{(n)}$ be the configuration at iteration n. The probability of a transition from $x^{(n)}$ to an accessible configuration $x^{(n+1)}$ is calculated as follows. Define the weight-normalized energy change $\Delta E(y) = \big(E(y) - E(x^{(n)})\big) / \sum_t w_t$ for a transition to configuration y, so that a decrease in energy gives $\Delta E < 0$. Dividing by the sum of the energy weights expresses

β per unit of weight-normalized energy, so that its values remain comparable when the weights are rescaled. Let $\mathcal{A}(x^{(n)})$ denote the set of configurations accessible from the current configuration, and let $Z_n = \sum_{y \in \mathcal{A}(x^{(n)})} e^{-\beta \Delta E(y)}$ be the partition function that normalizes probabilities over that set.

$$p\left(x^{(n)} \to x^{(n+1)}\right) = \frac{e^{-\beta \Delta E}}{Z_n}. \tag{16}$$

When the temperature is raised, the system becomes more likely to generate alternative configurations. Fig. 10 below shows the frequencies of final configurations of the form *pika* (/pɪka/) for four different values of β. As temperature increases (β decreases), non-standard configurations are generated more often.

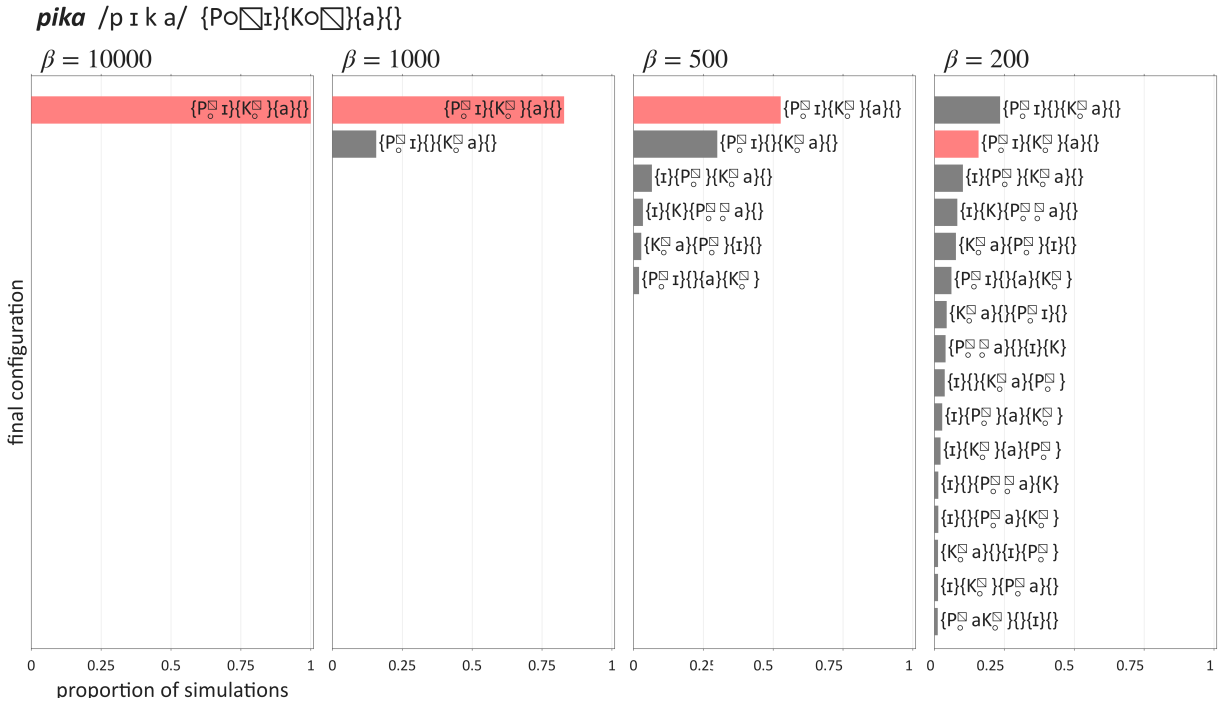


Figure 10: Effects of temperature on final configuration. Bars in each panel show final configuration proportions from 1000 simulation runs for a particular value of β. Red bars are the expected/canonical configuration. Final configurations that occurred on less than 1% of simulations are omitted for clarity.

Fig. 11 shows transitions in system state space trajectories from 1000 generated configurations with $\beta = 500$. Inspection of these shows that canonical configurations can transition to non-standard ones at sufficiently high temperature. For example, the state {pɪ}{k}{a}{} (#97), which is the canonical configuration, sometimes transitioned to {pɪ}{}{ka}{} (#95). This is notable because, in conventional terminology, {pɪ}{k}{a}{} corresponds to a coda-syllabified (or ambisyllabic) consonant and {pɪ}{}{ka}{} corresponds to a onset-syllabified consonant. The onset-syllabified /k/ (#95) notably also arose from trajectories that did not pass through the canonical state (#97).

The stochasticity in mode coupling is an important source of variation in configurations, and also a source of some speech errors. As shown in Fig. 10, as temperature

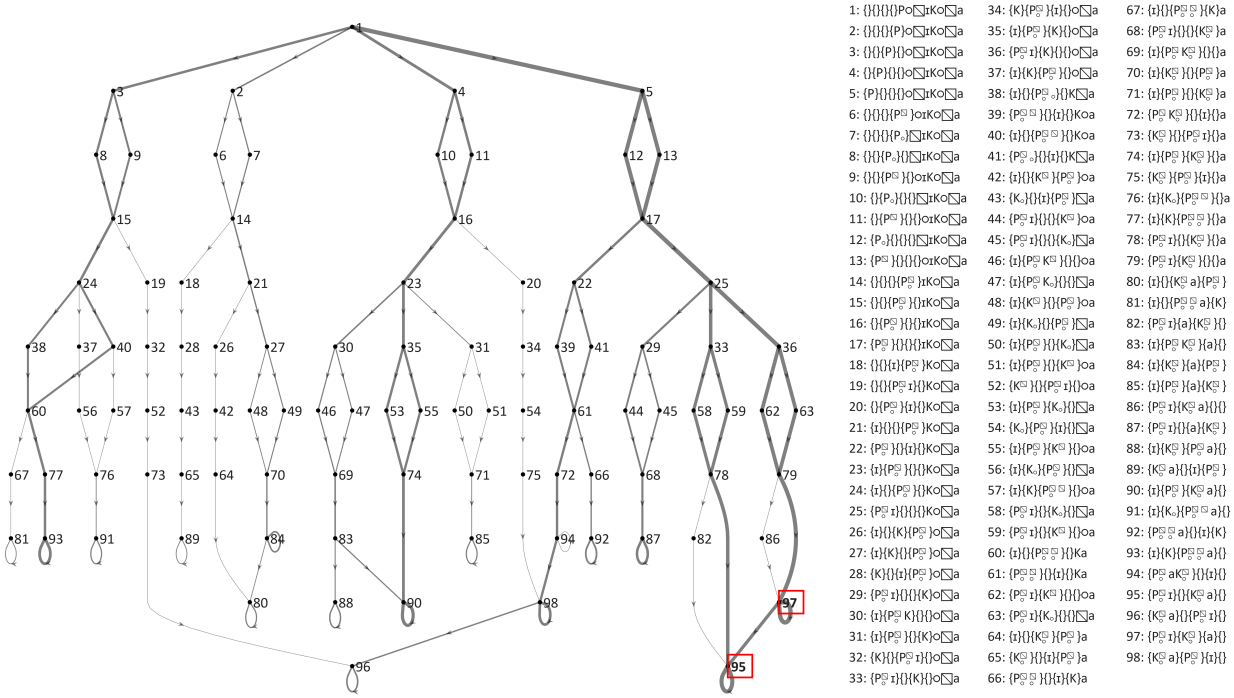


Figure 11: State space trajectories from Fig. 10 with $\beta = 500$, over 1000 simulations. States are labeled with numbers in the graph, corresponding to the labels on the right, sorted by the average iteration at which the state occurred. Edge thickness represents the proportion of times a transition occurred.

increases, a greater variety of configurations is generated. In general, some of these may correspond to configurations that generate typical vocal tract state trajectories (via subsequent mode selection/vocal tract control), while others may correspond to atypical state trajectories (so-called "errors" or non-standard productions). Some of the non-standard configurations correspond to common error patterns, such as syllable transposition (ka.pɪ). In the case of non-standard configurations that are not recognized as common errors, it is possible that the system may detect such configurations as degenerate prior to or during mode selection, in which case, a configuration-monitoring system may re-initiate the mode-coupling process. Hence not all atypical configurations are necessarily expected to be observed in the form of speech errors.

Regarding circumstances where the stochastic dynamics allow for multiple configurations that are viewed as standard (and hence different vocal tract state trajectories that are not interpreted as "errors"), the model can predict differences in the relative frequencies of these variants. It should be possible to test model predictions regarding these relative frequencies of configuration variants, but this is complicated by the fact that those frequencies are determined by relative weights of terms in the configuration energy, along with temperature. Additional challenges to this endeavor involve the reliability of empirical data and the fact that relative likelihoods of variants are likely vary as a function of speech rate and other exogenous factors.

3 Interpretation of phonological patterns

There are multiple scales of cognitive and environmental system dynamics responsible for patterns, and accordingly, multiple scales on which patterns can be identified. The order-creation directly associated with mode coupling occurs on a relatively fast timescale within the cognitive system. Based on studies of prepared speech initiation latency (Sternberg et al., 1988; Sternberg et al., 1978), where we can assume that lexical access/retrieval processes are relatively uninvolved, the minimal time for a configuration to stabilize is plausibly around 50-100 ms, with about 20 ms per additional syllable (i.e. mode pair). Hence mode-coupling itself is a fairly rapid and localized process.

Many phonological patterns may be only indirectly associated with mode-coupling because they depend on relatively larger-scale mechanisms: learning, social interaction, and language experience. In a broader analysis of phonological patterns (Fig. 12), each cognitive system has an acoustic, somatic, and visual surroundings. Information regarding the somatic surroundings of any given cognitive system is not directly available to the other cognitive systems. Acoustic environmental information, on the other hand, is prototypically accessible both to the speaker who produced it, and to other speakers nearby. And yet, gestural systems influence their somatic surroundings more directly than the acoustic surroundings. This leads to the following challenge for language learners: to communicate through speech, they must learn conceptal-gestural activation ($\mathbf{a}^{\mathrm{CG}}$ in Fig. 12) that *indirectly* results in self-generated acoustic environmental patterns that are similar to patterns created by other speakers.

Microscopically this learning is plausibly a matter of tuning synaptic efficacy of projections from neural populations associated with sensorimotor experiences (conceptals) to premotor populations associated with articulatory actions (gesturals). The

acoustic environmental order that cognitive systems encounter is compared against the order they create, and differences between the two drive changes in $\mathbf{a}^{CG}$. This is effectively a longer timescale feedback loop that constrains the system. Crucially, the process depends on there being sufficient regularity in the mapping between gestural organization and acoustic patterns, so that the comparison of self-generated acoustic order and other-generated order can be used to tune $\mathbf{a}^{CG}$.

From this perspective, mode-coupling can be viewed as regularizing mechanism, functioning to oppose perturbative mechanisms that can hinder the acquisition of normative conceptal-gestural activation. One of these is the increase in articulatory overlap that results from reliance on internal feedback for mode selection. When mode selection operates with internal feedback, degating and suppression processes are potentially dissociated. This allows for greater overlap of articulatory movements, increasing the rate of information transmission. But the downside of overlap is it makes acoustic order less readily associated with articulatory actions themselves. In other words, with too much overlap, the actions of a speaker may not be perceptually recoverable by listeners (Chitoran et al., 2002). Other perturbative mechanisms include aerodynamic effects, biomechanical interactions between articulators, and nonlinearity in auditory transduction of acoustic information—all of which can obscure the mapping from acoustic order to articulatory action.

In the absence of the constraints that mode-coupling imposes on speech production, there would be much less order in the articulatory and acoustic environment, and this presumably would make the task of acquiring normative speech patterns more difficult. Many diachronic sound changes have been analyzed as a form of hypocorrection, where learners reinterpret perturbative effects as intentional (Ohala, 1993). Crucial to this theory of sound change is a mechanism that prevents all perturbative effects from being reinterpreted in this way. Mode-coupling is a viable candidate for this mechanism because it constrains the conceptal-gestural activation patterns that can result in stable configurations, thereby serving as regularizing force.

Below we briefly discuss various forms of organization that have played important roles in conventional, segmental/symbolic phonological theories. In each case we consider how mode-coupling theory can inform our understanding of the relevant organization.

3.1 Segmental organization

In symbolic phonological theories, speech is conceptualized as a temporal sequence of segments (or phones, or phonemes; generically, *units*). These are metaphorically object-like entities arranged in a linear space that maps to a temporal order. Segmental reasoning thereby applies a logic of spatial/temporal exclusion: two distinct units on the same horizontal plane cannot occupy the same space, and vertical connections between units conform to transitivity of ordering (hence association lines cannot cross). In some theories, segmental units are also understood as containers of features, atomic elements that specify properties of sounds (Chomsky & Halle, 1968; Jakobson et al., 1951). Most features refer to characteristics of articulatory actions associated with sounds, although some refer to acoustic characteristics or abstract properties. Alternatively, segments have been conceptualized as nodes of a hierarchical structure that dominate a graph structure of features (i.e., a connected objects structure; Clements, 1985).

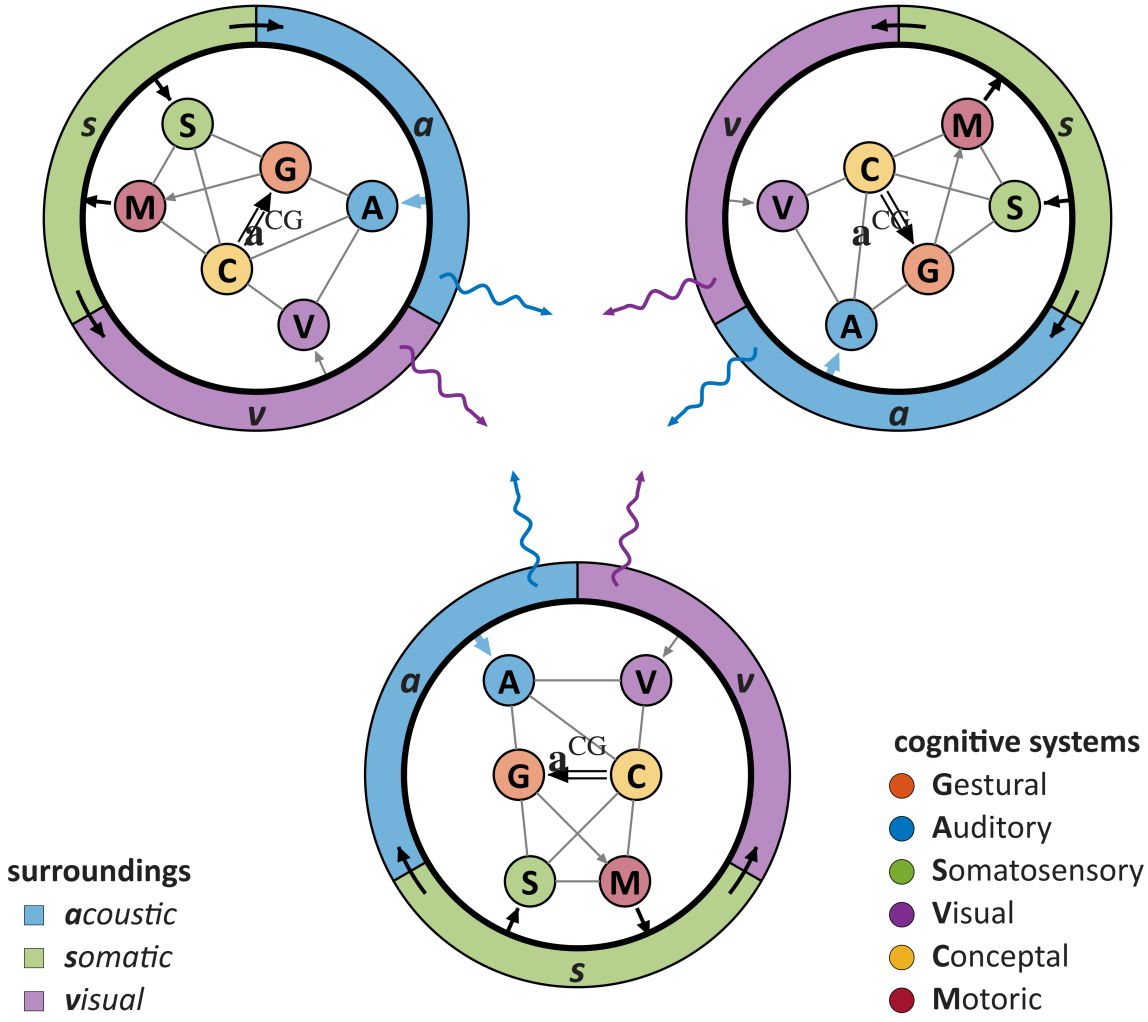


Figure 12: Schematic illustration of interactions between cognitive systems and their surroundings. Each of three cognitive systems (circles with black outlines) creates order in its somatic (s) and acoustic (a) surroundings through a motoric system (M) and transduces order from its acoustic (a), somatic (s) and visual (v) surroundings (rings) through sensory systems (A, S, V). Acoustic order created by other speakers is the primary source of information for learning conceptal-gestural activation ($\mathbf{a}^{\mathrm{CG}}$), an interaction between C and G.

The empirical problems with these theoretical constructs are far too numerous to discuss in detail here (see Tilsen, 2016). Many phoneticians (who tend to be more skeptical of phonological abstractions) have questioned the reality of segments (Browman & Goldstein, 1990; Ladefoged, 1984, 2001; Ohala, 1992; Port & Leary, 2005). Non-literate speakers and speakers in non-literature cultures perform worse than literate speakers on various phonemic awareness tasks (Morais, 2021; Morais & Kolinsky, 1994), suggesting that the hypothetical construct of speech segments (and phones, and phonemes) are artifacts of the relatively recent technological innovation of writing, rather than entities that are fundamental to phonological cognition. Likewise, language games in which segmental substitutions or transpositions occur are only found in literature cultures, suggesting that such patterns are mediated by orthophonographic mappings. The subset of speech errors that are commonly interpreted as substitutions or transpositions of segments can be alternatively interpreted as substitutions/transpositions of multiple features, and thus do not constitute evidence of segmental units. On top of this, it is suspicious that no phonological research has uncovered evidence for phonological patterns in which a count of phonemes or segments is necessary to describe the pattern—if segments were cognitively real units of organization, the most basic patterns one might expect to observe might involve counting.

The (somewhat questionable) utility of the hypothetical construct of a segment derives from the observation that somatotopically dissociated articulatory actions tend to co-occur in speech. For consonants, this co-occurrence involves (i) an action (or actions) that form a posture in the oral cavity (V- and C-class) and (ii) actions that govern the state of the velo-pharyngeal valve (N-class), the vibratory state of the vocal folds (G-class), protrusion/retraction of the lips (R-class), and somewhat less commonly, the elevation of the larynx (L-class). Mode-coupling theory accounts for these co-occurrences by positing that the former are i-type gesturals and the latter are d-type gesturals, subject to a dependency constraint. The dependency constraint plausibly derives from the fact that d-type gesturals are either unable to produce their prototypical acoustic effects in the absence of such co-occurrence, or that their prototypical environmental effects are transparently present or absent. The somatotopic systematicity of action co-occurrence that has been pervasively associated with “segments” or “phones” follows from this constraint, rather than the existence of segmental entities.

Furthermore, segmental theories have needed to distinguish between segments that are simplex and those that are complex/contour segments, because the latter—which typically involve multiple oral constrictions with asynchronous movement initiation (e.g. affricates [t͡ʃ], liquids ([l] and [ɹ]), diphthongs [ai], vowels with off-glides [ei], etc.)—are not amenable to the common object-schematic metaphor that phonemes are containers of uniquely-specified identifying properties (i.e. features). In mode-coupling theory, the distinction is unnecessary because there is no presupposition that i-type gesturals coupled to the same mode should form a coherent entity in the first place. Indeed, the theory readily accounts for the empirically observed asynchrony of movement initiation: exclusion disfavors multiple oral constriction or vocalic gestural excitations from occupying the same orbit. As shown in Fig. 13, the [TT clo] and [Tb nar] gesturals of the affricate (a) and the [TD clo] and [TB j] gesturals of the palatal on-glide (b) differ in phase; in contrast, the d-type labial protrusion [LP round] of a labialized stop can occupy that same orbit as a C-class

gestural. Diphthongs (d), long vowels (e), and vowels with off-glides (f) all involve i-type gesturals coupled to different modes, and hence exhibit initiation asynchronies due to they competitive dynamics of mode selection. Furthermore, some classes of complex segments are markedly different depending on how they are organized within the syllable. For example, the English lateral /l/ in coda position is typically produced with asynchronous tongue dorsum and lateral tongue tip gestures (in the model, a (−)-sign [ʌ] and [TT l]), but pre-vocalically only the lateral gesture occurs. The asynchrony is ultimately a consequence of exclusion constraints.

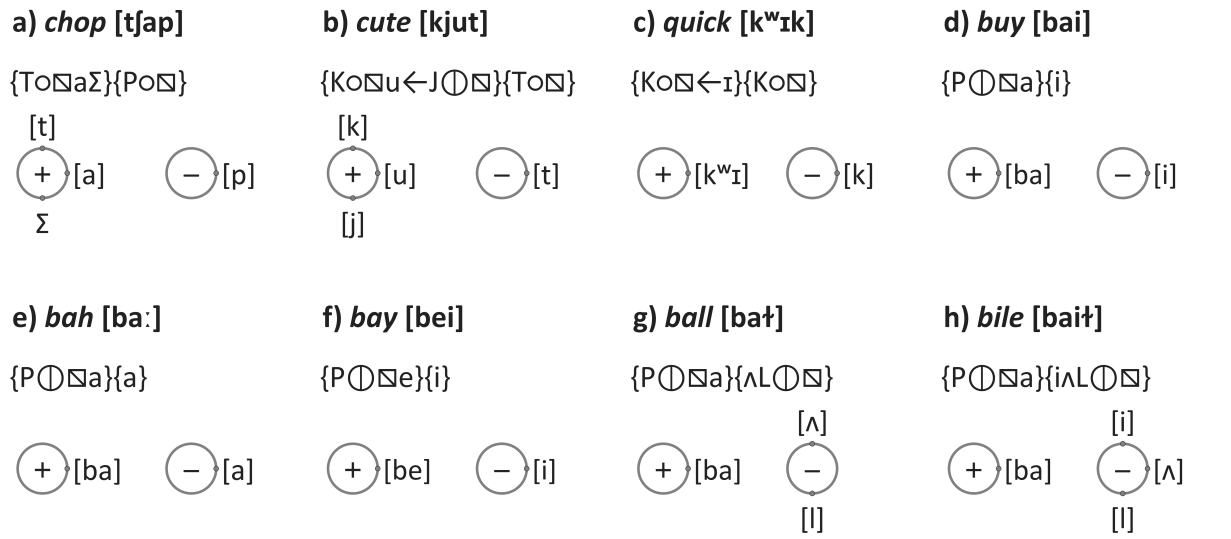


Figure 13: Configurations of forms with complex/contour segments. (a) affricate; (b) palatal on-glide; (c) labialized stop; (d) diphthong; (e) long vowel; (f) off-glide; (g) liquid coda; (h) diphthong with liquid coda.

From the perspective of mode-coupling theory, the absence of compelling evidence for segmental organization is expected. The prevalence of segments in traditional theories is a consequence of the fact that alphabetic writing systems are used for scientific communication, and this has led mainstream theories to reify graphemes as cognitive entities. Segments may be useful to a limited extent, but only because the constraints that govern mode-coupling apply in somatotopically motivated ways.

3.2 Moraic and syllabic organization

Object-based phonological theories have employed a variety of models of the internal structure of syllables, examples of which are shown in Fig. 14 (a-d). One of the most popular is (a) the onset-rime model (Selkirk, 1982), in which syllable nodes branch to onset and rime subconstituents, and the rime branches to a nucleus and coda. Some theories adopt the body-coda model (b) instead (Chen, 2011; Vennemann, 2011). Moraic models (c, d) are a common alternative, with variants in which onset nodes are strictly layered (Hyman, 1985) or may link to syllable nodes (Hayes, 1989). The notion that there is internal structure to syllables has also been disputed (Pierrehumbert & Nair, 1995). For the sake of parsimony, what is needed is a theory that

unifies our understanding of the phenomena that these various models have been designed to capture. The questions to ask are thus (i) why is it the case that syllables as hypothetical constructs are useful for describing phonological patterns, and (ii) how does mode-coupling theory conceptualize various phenomena that are described with subconstituent and moraic models?

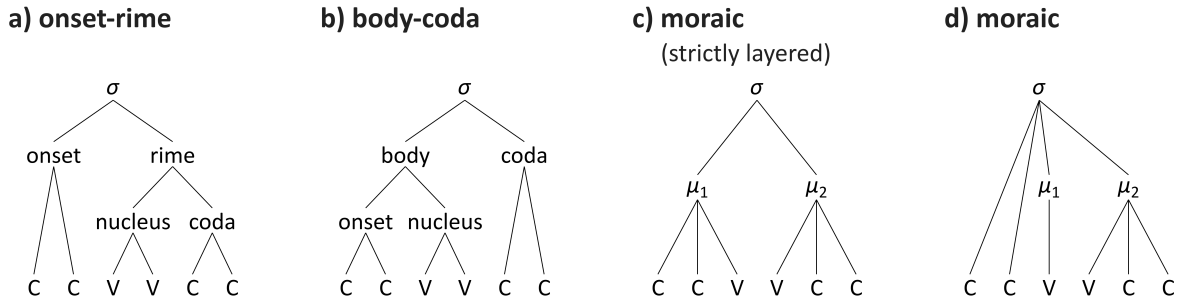


Figure 14: Models of syllable-internal structure. (a) onset-rime model; (b) body-coda model; (c) strictly layered moraic model; (d) moraic model with onset adjoined to syllable node.

In mainstream phonological reasoning, syllables are defined by a sonority peak—there is a particular segment in the syllable, often but not necessarily a vowel, that is the most "sonorous" one (roughly the loudest), and this most sonorous segment is often preceded and/or followed by less sonorous segments. Consequently, $(C_{-3}(C_{-2}(C_{-1}))$ $V(C_1(C_2(C_3)))$ are possible syllables, but VCV is not. In general the sonority of $C_{|n+1|}$ is greater than $C_{|n|}$: sonority tends to rise across prevocalic consonants and to fall across postvocalic consonants.

In mode-coupling theory sonority profile tendencies are epiphenomenal: the presence of a sonority peak associated with a vowel is a simple consequence of nucleus occupation: a {+/−} mode pair without a V-class gestural coupled to a {+} mode incurs a high energy cost, and coupling more than one V-class gestural to {+} is also costly. For diphthongs with a V-class gestural excitations coupled to both {+} and {−} modes, either V-class gestural may be the most sonorous, or they may be of equal sonority for long vowels. With mode coupling, there is no need to explain away apparent deviations, such as prevocalic sonority falls (e.g. fricative-stop sequences in *spy*, *stew*, etc.) or postvocalic sonority rises (e.g. stop-fricative sequences in *cats*, *naps*, etc.).

Phonological patterns that are traditionally analyzed in terms of syllabic organization arise from the circumstance that {+} and {−} coordination modes are grouped in pairs. Intuitions that speakers have regarding the number of syllables in a word form thereby correspond to the number of {+/−} pairs in its configuration. Syllabic organization of segments is generally held not to be encoded in long term memory, based on the observation that no languages appear to contrast words based on syllabification alone (Blevins, 1995). This suggests that syllable count intuitions require at least a motor simulation in which a gestural configuration is generated. For most word forms, speakers have the same count intuitions, but notably in English (and most likely in other languages) there are some forms for which intuitions vary between speakers or are inconsistent. Specifically, for word forms with a tense vowel or diphthong and liquid (r, l) coda such as *peel*, *hire*, *fool*, *file*, etc. some people have

the intuition that these words are a single syllable, others that they are two syllables, and still others express uncertainty (Lavoie & Cohn, 1999; Tilsen & Cohn, 2016). In a standard interpretation, the variation can be viewed as in Fig. 15: the liquid segment may be syllabified (a) as a coda, or (b) as the nucleus of a syllable. In mode-coupling, the distinction amounts to whether the [ʌ] gestural is associated with {−} or {+} sign: in the latter case, coupling to a second {+} mode is energetically favorable.

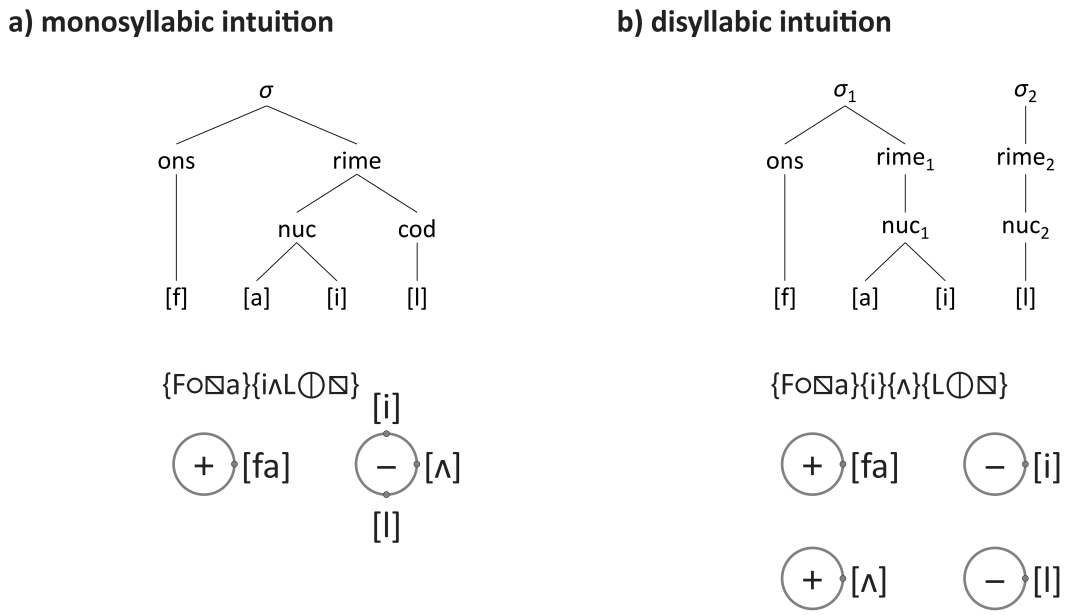


Figure 15: Interpretation of syllable count variation. (a) for monosyllabic *file* the V-class excitation has {−} sign and couples to {−} of the highest μ level, (b) for disyllabic *file*, the V-class gestural has {+} sign and is more likely to couple to the next highest level of modes.

The onset-rime model (Fig. 14 (a)) has been motivated by a number of empirical observations. First, the rime corresponds to a subset of intuitions regarding rhyming, i.e. so-called perfect rhymes (i.e. *cat* rhymes with *hat* but not *cot* or *cad*). This has been taken as evidence that the nucleus and coda form a unit (i.e. the rime) that speakers can assess for identity, to the exclusion of the onset. However, this particular intuition regarding rhyming is not cross-linguistically universal, and for multisyllabic forms with non-final stress, English rhyming intuitions incorporate the rime of a stressed syllable and all following segments. Thus *table* rhymes with *fable* but not *staple*. The language-specificity of rhyming and its extended scope in multisyllabic forms suggest that rhyming intuitions are not derived from an organizational pattern *per se* but rather are learned through experience of acoustic similarity in rhyming verse.

Second, there are a number of phonotactic asymmetries that correspond to onset-rime and nucleus-coda structural relations. For one, phonotactic prohibitions are more common between nuclei and coda (i.e. within the rime) than between onsets and nuclei. In English for instance there are many phonotactic constraints that apply to rimes, such as prohibitions against syllables with complex nuclei and non-coronal

complex codas (*/aimp/, */aiŋk/ vs. /ɪmp/), or prohibitions against monosyllabic words without a branching rime or nucleus (*/bɪ/ vs. /bɪt, bai/). In contrast, onset-nucleus phonotactic restrictions (the *body* unit of Fig. 14 (b)) are much less common across languages. These asymmetries in nucleus-coda vs. onset-nucleus combinatorics derive from a fundamental difference in how the gesturals associated with these units are organized. The gesturals of a body (onset and nucleus) are typically coupled to the same {+} mode, where exclusion is less restrictive: C-class gesturals repulsively phase-couple to one another but not with a V-class gestural. In contrast, some of the gesturals of a rime (nucleus and coda) are coupled to {−}, where all i-type gesturals, C- and V-class alike, are repulsively phase-coupled. All things being equal, this results in fewer available configurations for {−} modes.

Third, across languages it is observed that a single prevocalic segment (i.e. onset) is never prohibited but in some cases post-vocalic segments (codas) are restricted (e.g. Hawaiian has no codas and Japanese allows only nasal codas). As shown in Fig. 16, implicational hierarchies of syllable shapes inferred from studies of cross-linguistic typology (Zec, 2007) can be related to the energies of their configurations. All languages allow V and CV shapes, which are the two lowest-energy configurations. In contrast, some languages disallow VX (i.e. VV or VC), while others require VX (see discussion of minimal word requirements below). Implicational relations (arrows in Fig. 16 follow from the hypothesis that whenever a given orbit is allowed, lower-energy configurations that lack the orbit are also allowed.

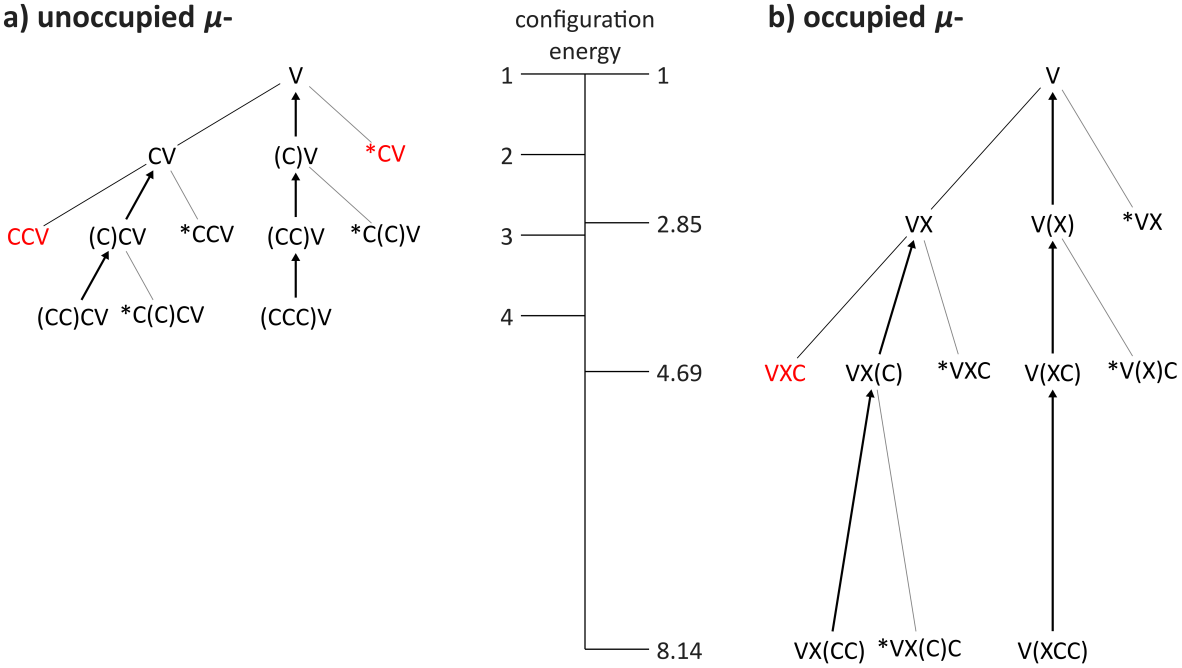


Figure 16: Schematization of the cross-linguistic typology of syllable shape phonotactics with implicational relations. Prohibited shapes are indicated by ('*'), and the absence of parentheses indicates an obligatory component of the shape. Red nodes are prohibitions/requirements that are unattested.

Other sorts of phonological patterns are more readily described by moraic models

(Fig. 14 (c, d)), which provide a notion of syllable "weight" by distinguishing between monomoraic ("light") syllables and bimoraic ("heavy") syllables. Patterns that have been analyzed with moraic structure include *minimal word requirements*: content words must be comprised of at least two moras, either a single heavy syllable or two light syllables; *quantity sensitive stress*: heavy syllables must have stress; and *contour tone restrictions*: contour tones (i.e. lexical tones that involve two or more F0 targets) can only occur on heavy syllables.

Not all languages exhibit weight-related patterns, but in those that do, moraic structural analyses posit that a long vowel or diphthong is linked to two moras within the syllable, as in Fig. 17 (a), rendering it heavy. In some languages, coda consonants are appear to contribute to weight (b), in others, they do not (b′). This variation in coda-weight can also be consonant-specific and can depend on position in word, with word-final syllables often differing from non-final ones. Minimal word requirements are satisfied by heavy monosyllables such as (a) or (b), or by disyllables (c), but are violated by light monosyllabic words (e). For example, in English heavy syllable content words like *pie* /pai/ or *pea* /pi:/ (CVV) and *pit* /pɪt/ (CVC) are common, as are disyllabic words like *pity* (CVCV), but monosyllabic words with a lax vowel like */pɪ/ or */pɛ/ (CV) are rare and mostly restricted to truncations or function words/discourse markers.

In mode-coupling theory syllable weight is reinterpreted in terms of gestural coupling to a {−}, but whether this manifests as weight can depend on the classes of those gesturals. In cases where codas are moraic, both V- and C-class gesturals coupled to {−} count for weight (as in Fig. 17 (b)). In languages where codas are non-moraic, only V-class gesturals count (Fig. 17 (b′)). Minimal word requirements can in both cases be interpreted as requirements that words must involve the selection of at least two modes: these are either the {+} and {−} modes of a heavy syllable, or two {+} modes, as in Fig. 17 (c). Similarly, contour tone restrictions are patterns in which a sequence of tone gestures must be associated with a {+/−} mode pair in which the {−} mode is occupied by an appropriate gestural. Such patterns can be viewed as the consequence of an exclusion principle between tone gestures along with language-specific variation in the dependency of tone gestures: in languages where tones are d-type gesturals, it follows that {−} must be occupied by some other gestural, hence the restriction.

Some moraic analyses also posit trimoraic ("superheavy") syllables, as in Fig. 17 (d). However, such analyses are often motivated only by the assumption that some particular consonants are necessarily moraic when post-vocalic, and it is not clear that there are phonological patterns that motivate a distinction between trimoraic and bimoraic syllables. Indeed, purportedly trimoraic syllables are often restricted to be word-final (Hall, 2000). This pattern follows from the relatively high energetic cost of an unbound gestural excitation in combination with an absence of a lower-level {+} mode: with no such mode present, word-final gesturals must couple to the lowest {−}, even if the configuration is energetically disfavored.

A fundamental concern in object-based theories is how segments associate with syllables. An often-invoked general explanation is the *maximal onset principle*, which holds that there is a preference for associating intervocalic consonants as onsets rather than codas. In the absence of other factors, this results in the structure of Fig. 18 (a), i.e. (C)V.CV rather than (C)VC.V. However, it has been argued in some languages, in order to satisfy a constraint that stressed syllables be heavy, the in-

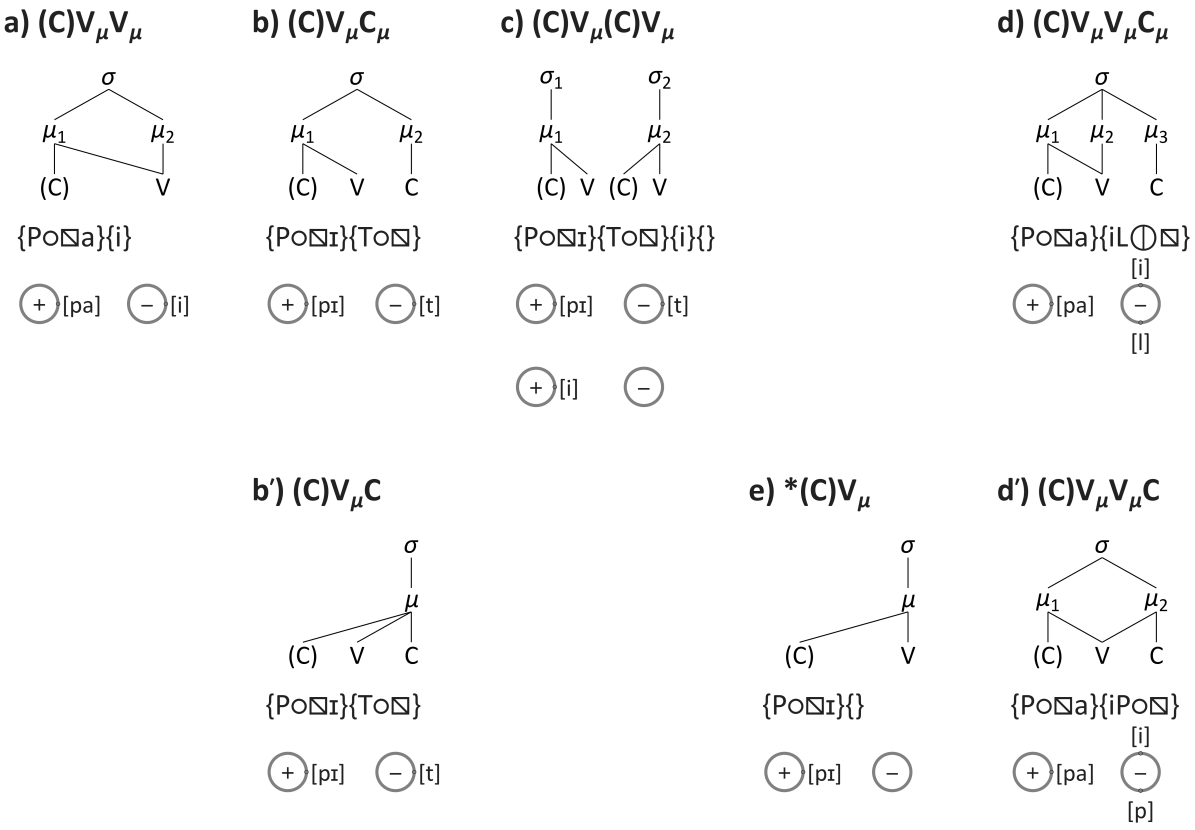


Figure 17: Moraic structural representations for weight-related phonological patterns. (a) long vowel. (b, b′) weight-bearing and non-weight-bearing codas. (c) disyllable. (d, d′) trimoraic syllable and bimoraic syllable with a non-weight-bearing coda. (e) violation of minimal word requirement.

tervocalic consonant is *ambisyllabic*—associated both with the coda of the preceding syllable and the onset of the following one, as in Fig. 18 (b). Mode-coupling theory interprets this as the consequence of coupling the intervocalic consonantal gestural to $\{-\}_\ell$ rather than $\{+\}_{\ell+1}$; articulatory studies generally support this interpretation. Relatedly, intervocalic geminate consonants are hypothesized to involve coupling of distinct excitations of a consonantal gestural to both $\{-\}_\ell$ and $\{+\}_{\ell+1}$ (Fig. 18 (c)). Thus mode-coupling offers an alternative analysis of syllabification patterns, without employing object connection metaphors.

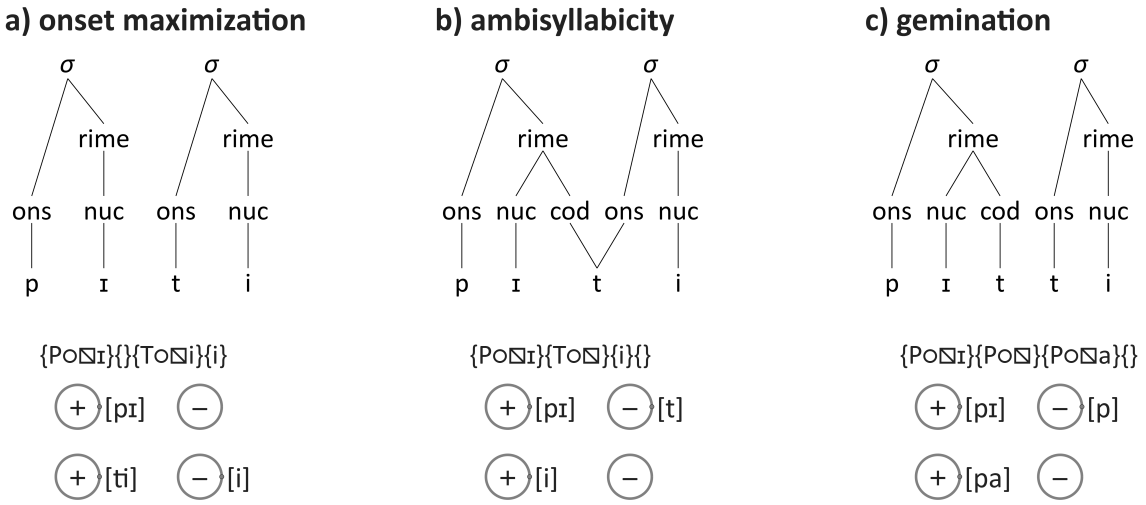


Figure 18: Syllabification patterns. (a) onset maximization; (b) ambisyllabicity; (c) gemination.

From a mode-coupling perspective, syllables are useful for describing phonological patterns because they correspond to {+/−} mode pairs, and moras are useful because they correspond to individual {+} and {−} modes. Cross-linguistic variation in the extent to which syllables or moras are appropriate for describing phonological patterns, rather than being a consequence of a configurational differences, may derive from differences in mode selection. Specifically, recall that internal feedback control allows for degating of μ_{m+1} before suppression of μ_m. Moraic vs. syllabic bias amounts to whether this asynchrony differs between modes within a level $\{+\}_\ell\{-\}_\ell$ and between levels $\{-\}_\ell\{+\}_{\ell+1}$. In the current model implementation this asynchrony is set by degating thresholds, the feedback level of the selected mode at which the next mode is released: λ_- for transitions within a level and λ_+ for transitions between levels, with a lower threshold giving greater asynchrony (the defaults, $\lambda_- = 0.42$ and $\lambda_+ = 0.20$, make transitions between levels the more asynchronous). For moraic bias, degating and suppression are similarly asynchronous. For syllabic bias, degating and suppression are relatively more asynchronous within levels than between levels; consequently syllabic-bias languages exhibit greater overlap within syllables than between them.

3.3 Metrical organization

In many languages, syllables in multisyllabic word forms are analyzed with a structural property called *stress*, and stress patterns appear to reflect an abstract grouping of syllables in units called *feet*, a so-called metrical organization in which stressed and unstressed syllables tend to alternate. Syllable stress is often asserted to be associated with acoustic properties of greater acoustic intensity, longer duration, and more extreme pitch movement, along with vowel reduction. However, the only cross-linguistically reliable diagnostic for stress is the ability for a syllable to bear pitch accent (see Tilsen, 2019b for discussion of stress diagnostics in Hayes, 1995). Theories of metrical organization are complicated by the fact that in addition to having a default pattern of stress that is systematically predictable from the number of syllables in a word form (sometimes called *fixed stress*), some languages also allow for stress to be lexically specified (called *free stress*) and/or conditioned on the moraic composition of syllables (quantity sensitivity, or "weight-based stress").

Recall from the description of metrical activation dynamics that when only an end-input is present, the default stress pattern is anchored to the lowest-level mode pair, and when only a begin-input is present, the default pattern is anchored to the highest-level pair. When the end-input is positive/negative, the pattern is iambic/trochaic; the reverse holds for begin-input—these patterns are a consequence of the inhibitory interactions between syllables on adjacent levels. The end-/begin-input parameters thereby allow for the system to capture a range of patterns observed in metrical typology (see Hayes, 1995). For English, which has a default metrical pattern of trochees from word end, $I^e = -1$ and $I^b = 0$. Examples of this pattern are shown in Fig. 19 (a) *ba.'nan.a* and (b) *ba.,nan.a.'ram.a*.

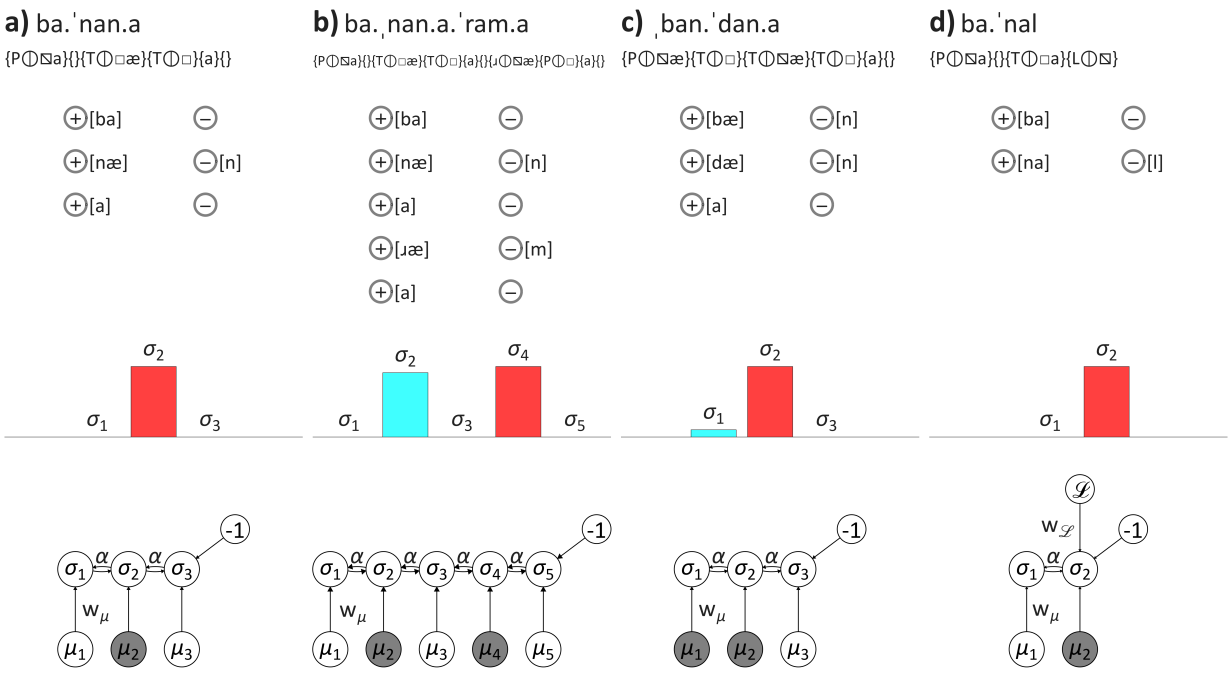


Figure 19: Examples of metricalization. Top row: mode configurations. Middle row: metrical activation values (baseline = 0); Bottom row: schematic of inputs and neighboring mode influences.

Quantity sensitivity arises from internal input (I^{μ^-}). This input increases the activation of a σ-system when its associated $\{-\}$ mode is occupied. Hence in Fig. 19 (c), the occupation of μ_1^- results in internal input to σ_1, leading to positive activation of σ_1. This variety of secondary stress notably arises in the course of the mode-coupling process, rather than being pre-determined.

The interplay between metrical activation and mode coupling can also work in the other direction, where metrical activation creates biases on configurations. For instance, intervocalic consonantal gestures are favored to couple to μ_ℓ^- rather than $\mu_{\ell+1}^+$ when σ_ℓ has sufficient metrical activation. This is a consequence of binding energy being a function of both mode level and metrical activation. Thus intervocalic consonantal gestural excitations couple to the stressed syllables of *banana* and *bananarama*, exhibiting an ambisyllabic pattern. Although occupied lowest-level $\{-\}$ modes contribute internal input like all occupied $\{-\}$ modes, in languages like English a negative end-input ($I^e = -1$) outweighs it ($2/3$). This relates to a pattern called extrametricality/extraprosodicity, in which the final syllables of words do not induce stress, regardless of whether they are heavy (e.g. *donor*, *tonal*). Another influence on stress patterns is lexical input, as with *banal* in Fig. 19 (d): here the iambic stress pattern arises due to additional information in the conceptal-gestural activation. In the current model this effect is implemented by associating a particular V-class gestural excitation with an abstract "accentual gesture". As shown for *banal*, presence of this accentual gesture in an orbit of a μ_2^+ substantially augments the activation of σ_2 and thereby suppresses activation of σ_1.

3.4 Prosodic word organization

Beyond metrical organization, various theories have argued for a higher-level unit of phonological organization, called a *prosodic word* (or phonological word, often symbolized with ω). However, unlike the patterns of organization discussed above (segmental, moraic, syllabic, and metrical), prosodic word organization is generally acknowledged to be quite flexible. For this reason, the prosodic word has been argued by some to be an emergent (as opposed to universal) form of organization (Schiering et al., 2010). In mode-coupling theory, prosodic word organization arises from constraints on conceptal configurations. These constraints limit the number of conceptals that can be simultaneously selected, and this in turn indirectly limits the number of gestural excitations that can occur in a configuration.

Indeed, many of the same principles that govern gestural configurations likely apply to conceptual-syntactic organization. This idea was discussed extensively in Tilsen (2019c). The basic premise is conceptal systems are activated by influences from their surroundings and from other conceptals. In preparing to speak, conceptal systems couple to syntactic coordination modes. Once the syntactic-conceptal configuration has stabilized, competitive selection of syntactic coordination modes results in a series of gestural configurations (prosodic words). One important consequence of this is that morphophonological patterns must occur within a prosodic word (i.e. within a gestural configuration).

For example, consider an utterance such as *the taxi is here*. The conceptals associated with this utterance can be organized for selection in a variety of ways, shown in Fig. 20. In (a), the definite determiner [DEF] and [TAXI] are selected together, while the verbal conceptal [BE] is selected with the predicate adjective [HERE]. In (b), [BE] is selected with [DEF] and [TAXI]. Note that in these two cases, the gestural excitations of [BE] differ: in the latter, it is a contraction. Morphophonological patterns of this sort result from modulation of conceptal-gestural activation conditioned on conceptual-syntactic configurations: the activation associated with [BE] differs depending on the conceptals it is co-selected with. Conceptal-gestural activation commonly varies in this way for grammatical conceptals such as those involving number (singular vs. plural) and person (1st, 2nd, or 3rd; e.g. I *am*, we/you/they *are*).

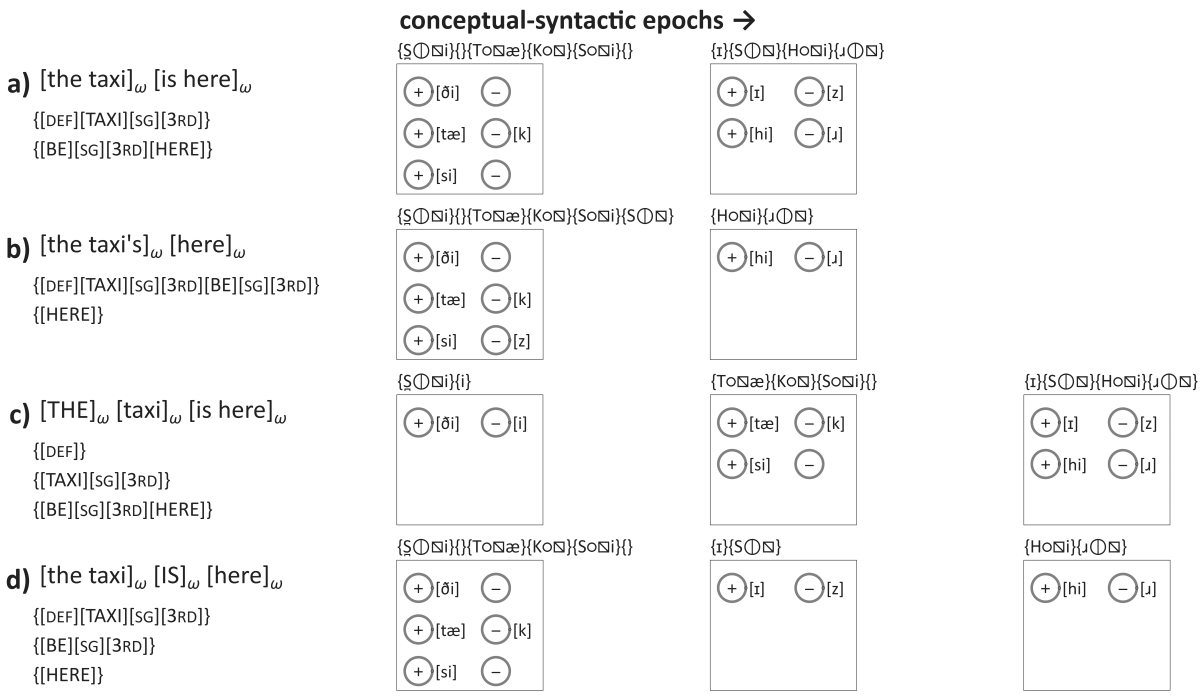


Figure 20: Prosodic word organization. Left: four different conceptal-syntactic organizations for the utterance *the taxi is here*. Right: gestural configurations in each epoch of conceptual-syntactic selection.

When a set of conceptal systems is selected, they induce the formation of gestural configurations through mode coupling. Gestural excitations associated with conceptals that are not in the currently selected set generally do not couple to coordination modes. Once the gestural configuration stabilizes, mode selection sequentially heightens gestural influences on vocal tract control fields. In the next conceptual-syntactic epoch, a subsequent set of conceptals is selected, the activation vectors associated with those conceptals give rise to a new gestural configuration and a new vocal tract state trajectory, and so on. Speakers can communicate pragmatic information, such as emphatic or contrastive focus, by organizing conceptals in different ways. For example, the utterance in (c) *THE taxi is here*, associates focus with the determiner by organizing it into a distinct set; likewise in (d) *the taxi IS here*, focus is on the verb. It is beyond the scope of this paper to examine principles of conceptual-syntactic organization in detail, but an important hypothesis is that many of the same factors involved in gestural organization via mode-coupling operate in conceptal organization (the parallel domains hypothesis; Tilsen, 2019c). The key point is that prosodic word organization is flexible because it reflects how conceptals are organized for selection.

3.5 Temporal locality

In standard phonological reasoning, a distinction is often drawn between local vs. non-local phonological patterns. This distinction is fundamentally a metaphoric one, based on the image-schematic presupposition that phonological organization is a spa-

tial arrangement of objects, where linear space corresponds to time. Patterns involving objects (often segmental units) that are not spatially adjacent are described as non-local. In order to impose locality where segmental schemas do not naturally allow for it, a number of approaches have proposed separate tiers for separate types of segments or for individual phonological features (Goldsmith, 1976; Heinz et al., 2011). Tiers are metaphorically parallel spatial coordinates. The motivation for this is patently the experiential intuition that only when entities are proximal in space can they interact.

It is generally held that there are two types of patterns that are apparently non-local: spreading harmonies and agreement harmonies (Hansson, 2001; Rose & Walker, 2011; Tilsen, 2019a). In spreading harmonies, some articulatory gesture appears to become highly active substantially earlier than otherwise expected (anticipatory or right-to-left R→L spreading), or alternatively remains highly active substantially later than otherwise expected (referred to as perseveratory or L→R spreading; see Gafos, 1999). For example, in some languages, nasality will spread in an anticipatory direction, so that the velum is lowered during sounds that precede a nasal consonant. This pattern always occurs within the domain of a prosodic word or is morpho-phonologically restricted, and it may be either a lexicalized characteristic of word forms, or arise when morphosyntactic combinations provide a suitable context.

Hypothetical examples in Fig. 21 (a,b) illustrate a spreading pattern. (a) shows that when a morphological root (phonemically /ba/) is suffixed with the form /sa/, which does not include a [VEL op] excitation, no anticipatory nasalization occurs, i.e. /ba+sa/ → [basa]. (b) shows that when the suffix activates [VEL op], nasalization "spreads" through orbits of preceding modes, i.e. /ba+na/ → [mãna].

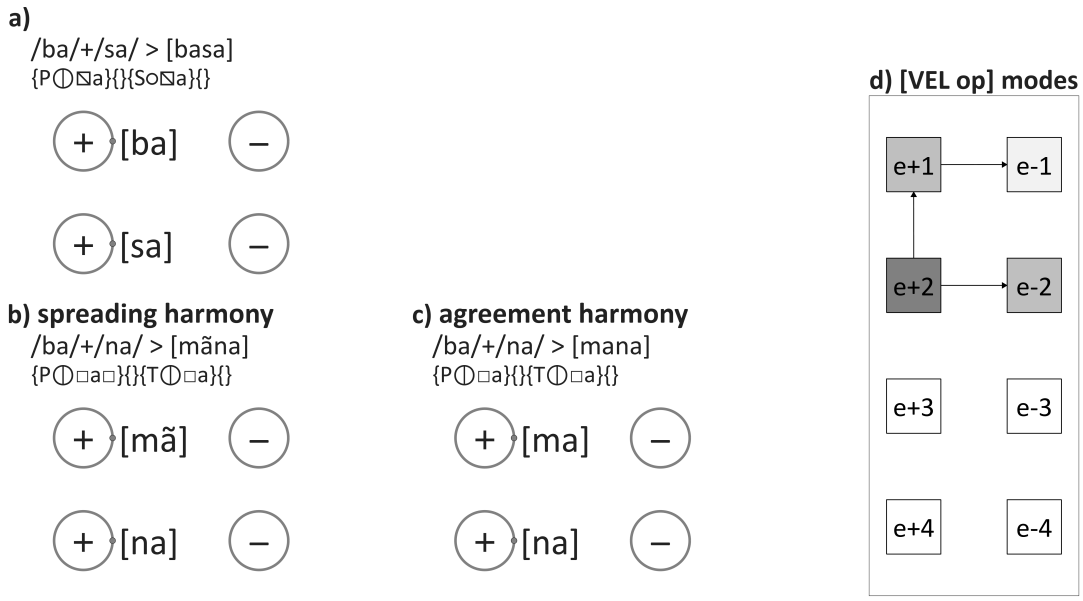


Figure 21: Harmonies as induction of excitation modes. (a) no harmony; (b) spreading harmony; (c) agreement harmony; (d) illustration of excitation induction.

A plausible mode-coupling interpretation of spreading harmony is that such pat-

terns are the result of interactions between gestural excitation modes. Note that these intragestural interactions have not yet been incorporated into the computational model, but they are nonetheless a reasonable consequence of the polychronous groups heuristic, whereby modes with distinct frequency and/or phase are microscopically associated with overlapping subpopulations of neurons. As shown in Fig. 21 (d), when a given conceptal (here the suffix associated with the form /na/) activates an internal excitation of a [VEL op] gestural, the active excitation could induce additional internal excitations of [VEL op] to become active and couple to other modes. Hence the [VEL op] excitations associated with spreading arise not directly from conceptal activation but from within-gestural interactions between modes. For empirical adequacy, it seems sufficient to constrain these *intragestural inductions* so that induced excitations are on the same level as the source excitation or a level adjacent to the source excitation. Induction to higher/lower levels leads to anticipatory/perseveratory spreading, respectively.

Spreading harmonies can also be "blocked", such that the presence of a segment appears to halt the spreading. For example, blocking occurs in /basa+na/ $\rightarrow$ [basãnã] due to the presence of the alveolar fricative [s]. The blocking pattern is likely a consequence of a phonotactic constraint that prohibits fricative gesturals from being nasalized. The phonotactic constraint itself derives from perceptual recoverability considerations: the oral pressure gradient required for generating perceptually salient turbulent airflow (i.e. a fricative sound) is compromised by venting of oral pressure through an open velo-pharyngeal valve ([VEL op]).

In segmental/featural terms, agreement harmonies differ from spreading in that segments intervening between the source and the harmonizing segments do not exhibit the articulatory correlates of the relevant feature. Furthermore, there are often constraints on featural properties of the harmonizing segment itself (e.g., stop consonants but not fricative consonants will harmonize). In the nasal consonant agreement harmony of Fig. 21 (c), only the consonantal gestural of the higher-level mode pair is nasalized. The same intragestural excitation induction mechanism that operates in spreading harmonies can account for this, with the additional stipulation that most of the induced excitations fail to couple to orbits between the source and target of the harmony—induced excitations will only couple to orbits occupied by some member of a particular subset of i-type gesturals. Thus in agreement patterns, excitations are induced with the same mechanism as in spreading, but only some of those excitations couple to μ-systems.

4 Discussion and conclusion

Mode coupling theory aims to provide a vocabulary for macroscopic description of phonological PATTERN and analysis of ORDER. It adopts a *biophysical* approach, meaning that its vocabulary coheres with vocabularies used in physical and biological reasoning. To accomplish this, it employs concepts from statistical physics to conceptualize both (i) how macroscopic states may be derived from microscopic ones, and (ii) how those states evolve in time. This is manifested in the act of partitioning the universe into cognitive systems and their surroundings; in further partitioning these into interacting subsystems; in construction of system state spaces; in specification of change rules that govern systems (as reduction of free energy); and in interactions

that comprise those change rules. These constructions are motivated to some extent from knowledge of the biological organism, particularly knowledge of (or speculation about) the organization and dynamics of its nervous system.

At the same time, we have used information theory to frame the empirical focus of mode-coupling theory as PATTERN, i.e., statistical dependence associated with reduced entropy. A natural question to ask is whether the physical and information vocabularies are compatible: is information equivalent to some coarse-grained form of negative entropy, or in other words, is information physical? A variety of perspectives exist on this general question (Jaynes, 1957; Landauer, 1991; Parrondo et al., 2015; Sethna, 2021). In mode-coupling theory specifically it is relevant to question whether the free energy of the model is merely metaphorically "energy" or whether it should be interpreted physically; and likewise, to ask whether the ORDER of interest is associated with a physically defined entropy. The physical connections are tenuous but not unimaginable, by reasoning as follows.

Suppose that the second law applies on the scale of an individual neuron. Any entropy reduction we can locally define on this scale will be accompanied by a greater increase of entropy in its surroundings (the complement of the local system). The "state" we define for a neuron (the local system) is necessarily a strategically coarse-grained macroscopic one. The second law extrapolation is possible because the coarse-graining by design ensures that directions of entropy change are preserved across scales. The next most reasonable extrapolation of the second law is to a population of neurons, where interactions between neurons must be accounted for. Suppose again that any local order creation that we can define on this scale is accompanied by a greater destruction of order. We again increase our scale of analysis and renormalize so that the system description preserves entropic principles. At this point it would be strange to decide that no further scale transformation is possible. Hence we can entertain the notion that there exists a macroscopic description of the cognitive state that conforms to the second law, because that description has been derived through recursive application of scale transformation with normalization. The same logic can be applied to obtain a free energy whose changes are consistent with those of our macroscopic entropy (in other words, $\Delta F = \Delta E - T\Delta S$ changes in the same direction across scales). Thus whether the "free energy" of the theory is physical depends on whether recursively applied transformations of this sort exist. If so, then it seems reasonable to draw parallels between how organisms make use of energy flows to create and maintain order within themselves and how conceptals/gesturals make use of metabolic energy flows to transition to highly ordered oscillatory states.

To assess mode-coupling theory, let us consider the biophysical desiderata of Section 1.2. Of these, neural compatibility is the most challenging to achieve, mostly because the nervous system is a high-dimensional complex system and because our capacities for observing and analyzing it are so limited. The motivations for the construction of interacting oscillatory systems—gesturals, their internal excitations, and coordination modes—are speculative but nonetheless motivated by the prevalence of oscillatory dynamics in the brain. Furthermore, the terms of configuration energy are motivated with varying degrees of speculation by reasoning about how those systems interact.

A notable shortcoming of the model presented here is that weights of configuration energy terms were optimized against a relatively moderate set of configurations.

This can plausibly be addressed by fitting the weights of configuration energy terms to a wider range of word forms. A more empirically oriented approach is ultimately desirable, in which weights are optimized by evaluating the vocal tract states they generate against empirical data. This endeavor is complicated by the fact that additional parameters associated with mode selection and vocal tract control would need to be jointly optimized with the weights of configuration energy terms. Moreover, the specific formulation of energy terms can be interrogated in revised. The current model adopts a set of terms which are sufficient to generate an empirically consistent range of patterns, but it is likely that the terms themselves can be simplified or formulated differently.

Another implementational shortcoming is the discretization of time in the generation of configurations. A more powerful and microscopically compatible approach would be to simulate the process with Langevin dynamics for continuous time, which would allow for more specific temporal predictions about the mode-coupling process to be made; this would also facilitate modeling phase and metrical activation dynamics in parallel with mode-coupling.

Because the nervous system is a physical system, some of the desiderata of Section 1.2 are effectively physical desiderata entailed by neural compatibility. Mode-coupling conforms well to these by design. For example, finiteness is manifested as the postulate that there is a limitation on the number of distinct gestural excitations that can be stable, which follows from the assumption that gestural excitation modes differ in frequency and/or phase; microscopically, the circuits responsible for the population oscillations have constrained ranges of oscillation frequency. Non-spatialization of time is accomplished by conceptualizing temporal sequencing as the consequence of a mode selection mechanism, which incorporates feedback systems into change rules for activation states. Never does this system directly encode time metaphorically as a spatial relation. Non-autonomy is embraced by the model vis-a-vis the role that conceptals have in activating gestural excitations. Memorylessness, on the other hand, has not yet been modeled in a satisfying way. Specifically, the heuristic rule used to determine conceptal-gestural activation (long-term memories of articulatory actions associated with word forms) is clearly *ad hoc*. A more desirable approach would be to model the relatively slower timescale learning dynamics that govern conceptal-gestural activation, in the context of multiple cognitive systems interacting over time. Nonetheless, in its current form the model is not inconsistent with memorylessness.

Regarding the desideratum of empiricality, Section 3 showed how a variety of forms of organization can be interpreted in the theory, addressing patterns that are conventionally associated with segmental, moraic/syllabic, metrical, and prosodic word units. Of course, there are far more phonological patterns that must be described by the theory in order to achieve empirical coverage; a comprehensive analysis is necessarily deferred for future work. It is nonetheless evident that mode-coupling theory has the analytical power to describe many patterns, and so it is plausible that the theory can provide a fully unified description.

Formal phonologists may see parallels between the use of a Boltzmann distribution for generating gestural configurations and various constraint-based models that either select a surface form via optimization, i.e. optimality theory (Prince & Smolensky, 2004) or assign probabilities to surface variants, i.e. harmony theory (Smolensky, 1986). Indeed, there is a non-trivial similarity in that mode-coupling, like constraint-

based models, is based on minimization of an energy-like quantity.

However, there are some consequential differences between these approaches. First, the energy calculations of mode-coupling are derived from speculative hypotheses/heuristic assumptions regarding the microscale dynamics of neural population interactions. Thus the model aims for compatibility between the microscale and macroscale. This contrasts with standard optimization-based frameworks, which define energies in terms of symbolic, object-schematic patterns, without much effort to motivate those energies microscopically. Second, the mode-coupling process is inherently dynamic (albeit discretized); in contrast, most optimization-based models do not generate trajectories of configurations, i.e. sequences of macroscopic states. The importance of temporal dynamicity is apparent when the configuration process is analyzed sufficiently far from the zero temperature limit. In this regime, stochastic influences on configuration state trajectories predict the possibility of alternative stable configurations and degenerate configurations. This feature allows the model to make more specific predictions regarding empirical patterns.

Perhaps the closest existing model to the one proposed here is Gradient Symbolic Computation (GSC) (Smolensky et al., 2014). The main similarity is that both models adopt a relaxation mechanism, in which an energy function defined over macrostates guides the evolution of the macrostate. A key difference is that the macrostates of mode-coupling are not viewed as symbols, but rather, as states of oscillatory systems. One might argue that the simplifications adopted in the implementation of the model effectively make it symbolic, since the macroscopic state variables of gestural modes $[r,\ \theta,\ f]$ are not modeled with a continuous dynamics. Under this interpretation, the "symbols" of the current model are markedly different from those of GSC, in regards to the logic of operations that governs their behavior. In GSC, the symbols are related "via the fundamental combinatory operation of symbolic computation, concatenation" (Smolensky et al., 2014). In contrast, the macroscale operations of mode-coupling theory are couplings/decouplings between intrinsically oscillatory systems. Moreover, mode-coupling theory adopts a simpler hypothesis regarding the types of symbolic entities on the macroscale: in GSC, symbols are members of a "multi-set of phonological segments related in a particular order and grouped into constituents such as syllables [σ], stress feet (Ft), and prosodic words [PrWd]" (Smolensky et al., 2014); in mode-coupling the macroscale entities ("symbols") are more simply coordination modes and gestural excitations; patterns conventionally associated with a hierarchy of constituent units are epiphenomena of constraints on how the macroscale entities interact, rather than being dimensions of the state space.

Instead of conceptualizing gestural excitations as symbols, I prefer to think of symbols and the constructs of mode-coupling theory as aspects of fundamentally different vocabularies for description of macrostates, requiring entirely different logics of change. Symbols are object-like and their spatial relations encode temporal ones; hence the logic of change is one of object manipulation and containment and/or attachment. Modes are states of oscillatory systems and their relations are couplings; hence the logic of change is a logic of synchronization. The configurations of mode-coupling theory are transiently stationary states, forms of local ORDER. Symbolic vocabularies may grossly approximate these transiently stationary states, but the vocabulary of mode-coupling aims to describe them in a way that can be put to test with future advances in neuroimaging.

Code and data availability

All code, input data and scripts needed to reproduce the simulations and figures in this paper are available at https://github.com/tilsen/mode-coupling.

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